

UNIT-I

ENGINEERING PHYSICS

UNIT-1 PART -A

INTERFERENCE

2 Marks Questions

1 Explain Principle of super position of waves:

When a medium is disturbed simultaneously by any number of waves, the instantaneous resultant displacement of the medium at every instant is the algebraic sum of the displacements of the medium due to individual waves in the absence of the other.

If $y_1, y_2, y_3, y_4, y_5, \dots$ be the displacement vectors due to waves 1,2,3,4.... Acting separately, then the resultant displacement 'y' is given by

$$y = y_1 + y_2 + y_3 + y_4 + \dots$$

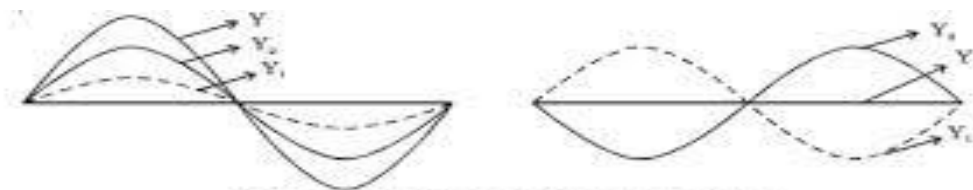


Fig 5.14 Superposition principle

This is known as principle of superposition. If $\vec{Y}_1$ and $\vec{Y}_2$ represent the individual displacement then the resultant displacement is given by
$$\vec{Y} = \vec{Y}_1 + \vec{Y}_2$$

Linear superposition of two waves of same frequency:

Let us consider two simple harmonic waves

$$y_1 = a_1 \sin \omega t \rightarrow (1)$$

$$y_2 = a_2 \sin (\omega t + \phi) \rightarrow (2)$$

where a_1, a_2 are amplitudes

ϕ is the phase difference acting on a medium simultaneously, then the resultant displacement is given by

$$y = y_1 + y_2 = a_1 \sin \omega t + a_2 \sin (\omega t + \phi)$$

$$= a_1 \sin \omega t + a_2 \sin \omega t \cos \phi + a_2 \cos \omega t \sin \phi$$

$$= \sin \omega t (a_1 + a_2 \cos \phi) + a_2 \cos \omega t \sin \phi$$

$$\text{Let } (a_1 + a_2 \cos \phi) = A \cos \theta \rightarrow (3)$$

$$a_2 \sin \phi = A \sin \theta \rightarrow (4) \quad \text{where } A \text{ and } \theta \text{ are constants}$$

$$y = A \sin(\omega t + \phi)$$

Squaring and adding equation (3) and (4)

$$(a_1 + a_2 \cos \phi)^2 + (a_2 \sin \phi)^2 = A^2$$

$$A = \sqrt{a_1^2 + a_2^2 + 2a_1 a_2 \cos \phi}$$

Thus, the resultant amplitude at any point depends on the phase difference of the two waves meeting at that point

Special cases:

$$\phi = 0 \rightarrow A = a_1 + a_2$$

$$\text{If } \phi = \pi \rightarrow A = a_1 - a_2$$

2. Explain the Phenomenon of Interference of light:

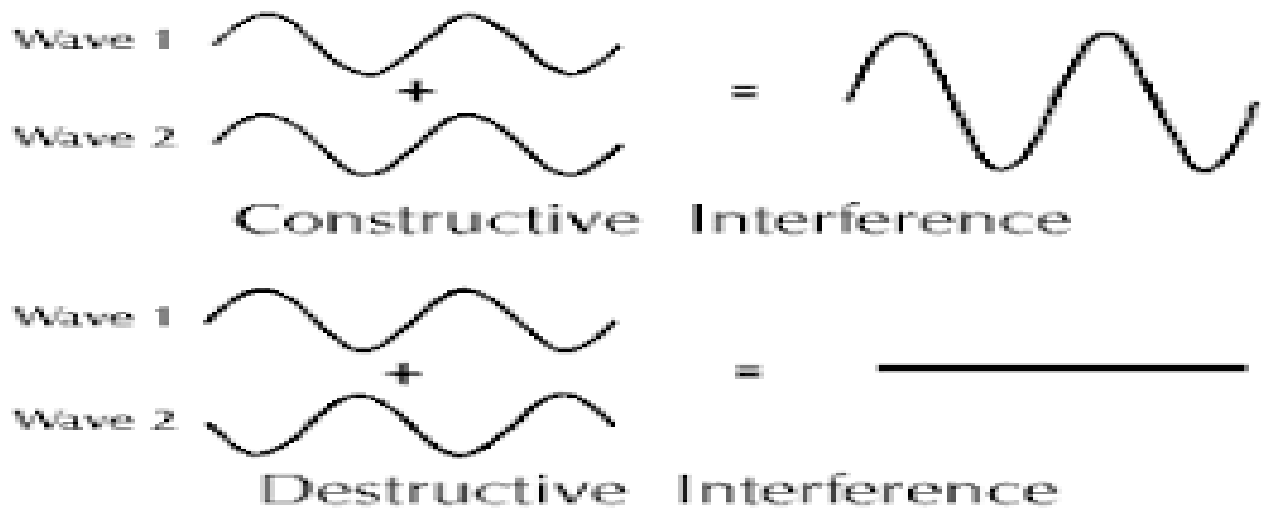
When two light waves of same frequency propagating in same direction and maintain constant phase difference are superimposed, then the resultant amplitude in the region of superposition is different from individual intensities. This modification in the distribution of intensity in the region of superposition is called interference.

Constructive interference:

The crest of one wave strikes the crest of another wave or the trough of one wave strikes the trough of another wave. This is the case of “Constructive interference”.

Destructive interference:

If the crest of one wave superimposes on the trough of other wave and vice-versa, this is the case of destructive interference.



3.What are the necessary Conditions for obtaining interference fringes

The sources should be monochromatic.

The sources must be narrow

The two sources should be coherent

The two sources must emit continuous waves of the same wavelength

The separation between the two sources should be small.

The distance between the source and screen should be large.

The back ground should be dark.

The amplitude of the interfering waves should be equal or nearly equal.

4. Explain concept of Coherence Source in Interference of light

When a light source emits a light wave with same frequency, same wavelength and constant phase difference that source is called coherence source.

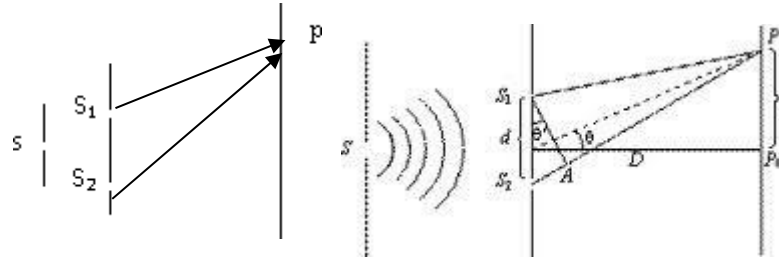
The coherence source is classified under two types

1.Temporal coherence

2.Special coherence

5. What are the conditions for maximum and minimum intensity

Consider a monochromatic source of light 'S' and 'S₁'
'S₂' be two narrow pinholes equidistant
from the source(S) shown in fig



Let a_1 , a_2 are amplitudes of two waves from 'S₁' and 'S₂'
and ϕ be the phase difference between the two waves
reaching at point 'P'. The path difference between two
waves arriving at P is $S_1 P - S_2 P$

$$\phi = \frac{2\pi}{\lambda} (S_1 P - S_2 P)$$

Condition for maximum intensity:

Intensity 'I' is maximum, when $\cos\phi = +1$ i.e, phase
difference $\phi = \pm 2n\pi$, where $n=0,1,2,3,\dots$

$$S_1 P - S_2 P = n\lambda \text{ (maximum)}$$

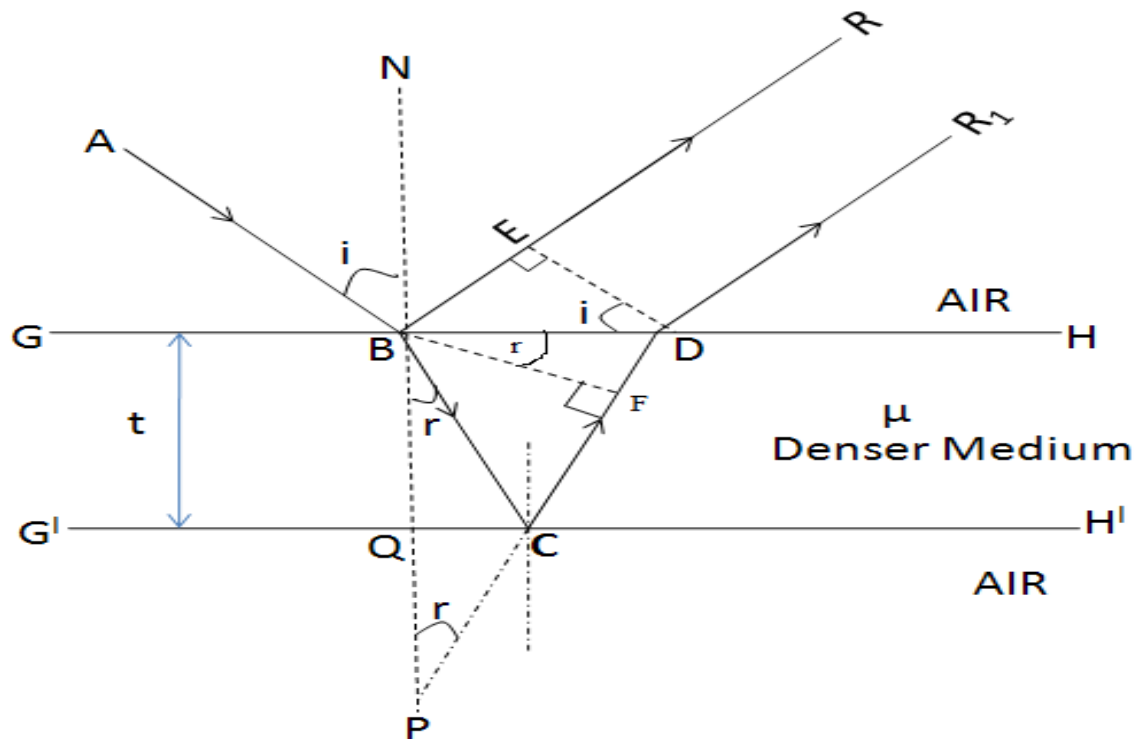
Condition for minimum intensity:

Intensity 'I' is minimum, when $\cos\phi = -1$ i.e, phase
difference $\phi = (2n+1)\pi$ where $n=0,1,2,3,\dots$

$$S_1 P - S_2 P = (2n+1)\frac{\lambda}{2} \text{ (minimum)}$$

10 Marks Questions

1. Derive an expression for the interference of light in thin films due to reflected light:



Let GH and G_1H_1 be the two surfaces of a transparent film of uniform thickness 't' and refractive index ' μ ' as shown in above fig.

Suppose a ray AB of monochromatic light be incident on its upper surface then the ray is partly reflected from lower and upper surfaces. These are in position to superimpose they can give the interference pattern. The optical path difference between the two reflected light rays (BR and DR_1) is given by

$$\Delta = \mu (BC + CD) - BE \rightarrow (1)$$

$$\text{We know } \mu = \frac{\sin i}{\sin r}$$

$$\Delta^{\text{le}} \text{ BED} \quad \sin i = BE/BD$$

$$\Delta^{\text{le}} \text{ BDF} \quad \sin r = DF/BD$$

$$\mu = \frac{\sin i}{\sin r} = \frac{BE/BD}{DF/BD} = \frac{BE}{DF}$$

$$\therefore BE = \mu(DF) \rightarrow (2)$$

$$\text{From eqn. (1)} \quad \Delta = \mu(BC+CD) - \mu(DF)$$

$$\Delta = \mu(BC+CD - DF)$$

$$= \mu(BC+CF+\cancel{DF} - \cancel{DF}) \quad (\because CD=CF+DF)$$

$$= \mu(BC+CF) \quad (\because BC=PC)$$

$$= \mu(PC+CF) \quad (\because PF=PC+CF)$$

$$= \mu(PF) \rightarrow (3)$$

$$\text{From } \Delta^{\text{le}} \quad BPF \cos r = \frac{PF}{BP} \quad (\because BP=2t)$$

$$PF = 2t \cos r \rightarrow (4)$$

Substitute eqn (4) in eqn(3)

we get $\Delta = 2\mu t \cos r$ This is usually called as **cosine law.**

We know that a ray reflected at a surface backed by denser medium suffers an abrupt phase change of π which is equivalent to a path difference $\frac{\lambda}{2}$.

Thus, the effective path difference between the two reflected rays is $\underline{2\mu t \cos r} + \frac{\lambda}{2}$.

2.Explain how Newton's rings are formed in reflected light.

When a Plano - convex lens with its convex surface is placed on a plane glass plate, an air film of gradually increasing thickness is formed between the two. If monochromatic light is allowed to fall normally, and the film is viewed in reflected light, alternately dark and bright rings concentric around the point of contact between the lens and glass plate are seen. The phenomena was first described by Newton i.e., why the rings are known as Newton Ring's by his name.

The experimental arrangement of obtaining Newton Ring's is shown in fig. 1

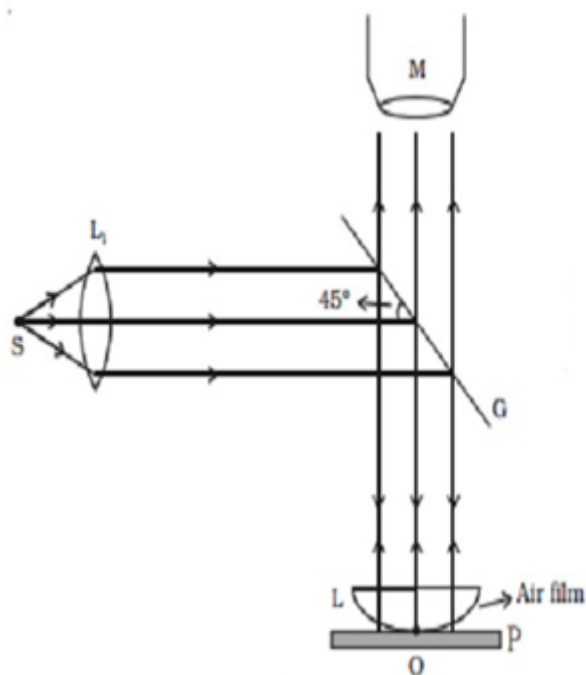


Fig 1 Newton's rings



Newtons rings

Newton Rings are formed due to interference between the waves reflected from the top and bottom surfaces of the air film formed between the lens & glass plate. These can be explained with the help of fig. 2.

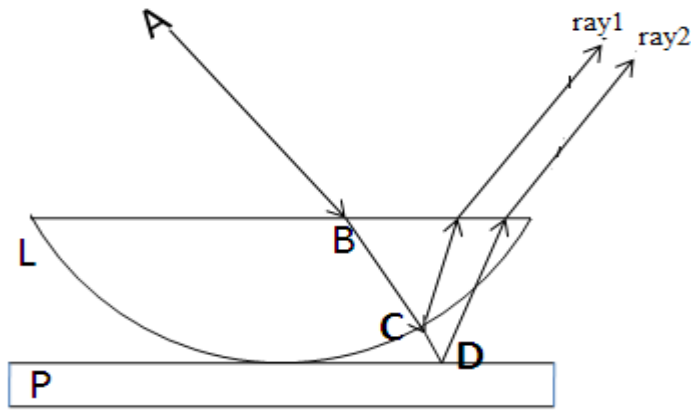


fig-2

AB is a monochromatic ray of light which falls on the system. A part is reflected at 'C' which goes out in the form of rays '1' without any phase reversal. The other part is reflected and goes out in the form of ray '2' with phase reversal of π . The reflected rays '1' and '2' are in a position to produce interference fringes as they have been derived from the same ray AB and hence fulfill the condition of interference. As the rings are observed in the reflected light, the path difference between two reflected rays (ray 1 & ray 2) is given by $2\mu t \cos r + \frac{\lambda}{2}$.

For air film $\mu=1$ and for normal incidence $r=0$,

hence in this case, the path difference is $2t + \frac{\lambda}{2}$.

Condition for maxima $2t + \frac{\lambda}{2} = n\lambda$

$$2t = n\lambda - \frac{\lambda}{2}$$

$$2t = 2n\frac{\lambda}{2} - \frac{\lambda}{2}$$

$$2t = (2n-1)\frac{\lambda}{2}$$

Condition for minima $2t + \frac{\lambda}{2} = (2n+1)\frac{\lambda}{2}$,

$$2t = (2n+1)\frac{\lambda}{2} - \frac{\lambda}{2}$$

$$2t = 2n\frac{\lambda}{2} + \frac{\lambda}{2} - \frac{\lambda}{2}$$

$$2t = n\lambda$$

3. Derive an Expression for diameter of dark and bright Newton's rings viewed in reflected light system.

Let LOL^1 be the lens placed on a glass plate. The curved surface LOL^1 is the part of spherical surface with centre at 'C'. Let R be the radius of curvature and 'r' be the radius of Newton's ring corresponding to the constant film thickness 't'.

For maxima i.e., for bright ring $2t + \frac{\lambda}{2} = n\lambda$

$$\Rightarrow 2t = (2n-1) \frac{\lambda}{2}$$

For minima i.e., for dark ring

$$2t + \frac{\lambda}{2} = (2n + 1) \frac{\lambda}{2}$$

$$2t = n\lambda$$

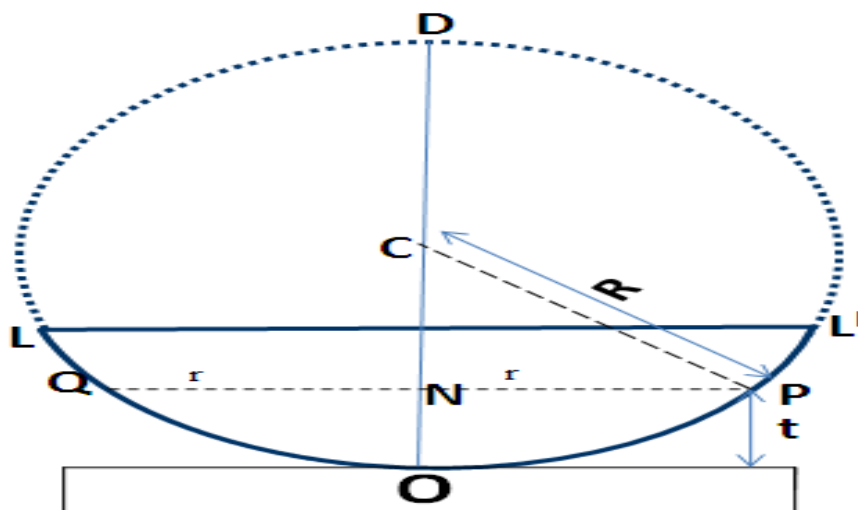


fig 3

From the property of the circle $NP \times NQ = NO \times ND$

$$r \times r = t \times (2R - t)$$

$$r^2 = 2Rt - t^2$$

$$r^2 \approx 2Rt$$

$$\therefore t = \frac{r^2}{2R}$$

Thus, for bright ring $r^2 = \frac{(2n-1)R\lambda}{2}$ $\left(\because r^2 = \frac{D^2}{4} \right)$

$$\left(\frac{D}{2} \right)^2 = \frac{(2n-1)R\lambda}{2}$$

$$D^2 = (2n-1)(2\lambda R)$$

$$\therefore D \propto \sqrt{(2n-1)}$$

Thus, the diameters of the bright rings are proportional to the square roots of the odd natural numbers.

Thus for dark ring $r^2 = n\lambda R \implies \left(\frac{D}{2} \right)^2 = n\lambda R$

$$\therefore D^2 = 4n\lambda R$$

$$D \propto \sqrt{n}$$

Thus, the diameters of the dark rings are proportional to the square roots of the natural numbers.

Determine the wave length of light λ

Let 'R' be the radius of curvature λ be the wavelength of light and n & m be the diameter of n^{th} & m^{th} rings respectively, then.

For the diameter of n^{th} dark ring we have

$$D_n^2 = 4n\lambda R \text{ ----- 1}$$

For the diameter of m^{th} dark ring we have

$$D_m^2 = 4m\lambda R \text{ ----- 2}$$

Eqn 2 - 1

$$D_m^2 - D_n^2 = 4m\lambda R - 4n\lambda R$$

$$\lambda = \frac{D_m^2 - D_n^2}{4R(m-n)}$$

UNIT-I

ENGINEERING PHYSICS

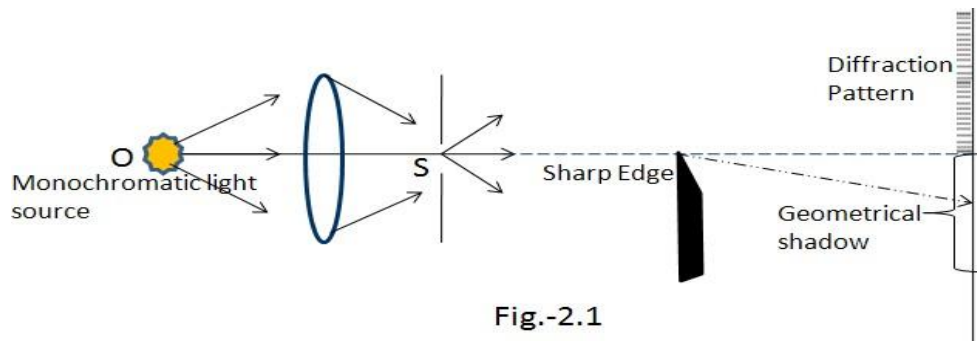
PART -B

DIFFRACTION

2 Marks Questions

1.What is meant by diffraction of light

When light falls on obstacles or small apertures whose size is comparable with the wave length of light, there is a departure from straight lines propagation, the light bends round the corners of the obstacle or apertures and enters in the geometrical shadow. This bending of light is called diffraction.



2.What are the Difference between Fresnel and Fraunhofer diffraction:

Fresnel	Fraunhofer
1) The distance between source and screen are at finite distance.	1) The distance between source and screen are at infinite distance
2) No lens is required to observed diffraction fringes	2) Two convex lenses are required to observed diffraction fringes
3) The diffraction effect can be observed at unidirectional	3) The diffraction effect can be observed at multidirectional
4) The source must be spherical waves	4) The source must be Plane waves
5)Mathematical investigation are complicated	5)Mathematical investigation are easy

3.what are the Difference between interference and diffraction:

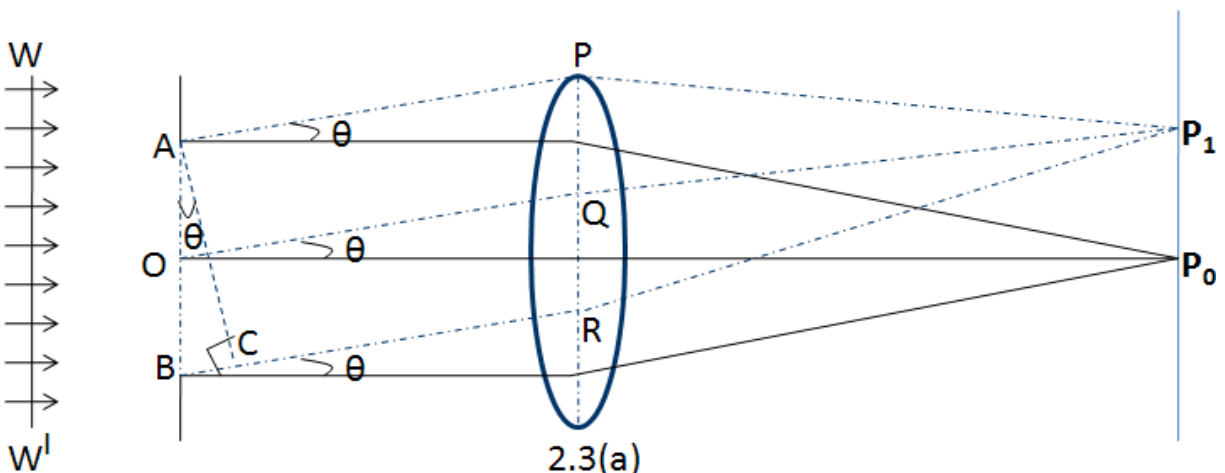
Interference	Diffraction
1) In this phenomenon interaction takes place between two separate wave fronts originating from two coherent sources.	1) Interaction takes place between secondary wavelets originating from different points of the exposed parts of same wave front.
2) Regions of minimum intensity are usually almost perfectly dark.	2) Regions of minimum intensity are not perfectly dark
3) The width of fringes may or may not be equal or uniform.	3) The width of fringes never equal
4) All the maxima are at same intensity	4)All the maxima are at varying intensity

10.Marks Questions

1.Discribe the formation of Fraunhofer's diffraction pattern due single slit:

Let a plane wave front WW^1 of monochromatic light of wave length ' λ ' propagating normally to the slit (AB) be Incident on it.

Let the diffracted light be focused by means of convex lens on a screen placed in to focal plane of the lens. According to Huygens-Fresnel, every point of the wave front in the plane of the slit is a source of secondary spherical



wave lets, which spread out to the right in all direction. The secondary wavelets travelling normally to the slit i.e, along the direction OP_0 , are brought to focus at P_0 by the lens. Thus P_0 is the bright central image.

The secondary wave lets travelling at an angle ' θ ' with the normal are focused at a point P_1 on the screen.

Path deference from Triangle ABC

$$\sin\theta = BC/AB$$

$$BC = AB \sin\theta \text{ here } AB = e \text{ (width of the slit)}$$

$$BC = e \sin\theta \text{ this is known as path deference}$$

$$\therefore \text{Corresponding phase difference} = \frac{2\pi}{\lambda} (e \sin\theta)$$

Let us consider that the width of the slit is divided into 'n' equal parts and the amplitude of the wave from each part is 'a'. The phase difference between any two consecutive waves from these parts would be $\frac{1}{n}$ (Total phase)

$$= \frac{1}{n} \left(\frac{2\pi}{\lambda} e \sin\theta \right) = d(\text{say})$$

Resultant amplitudes (R) is given by

$$R = a \frac{\sin \frac{nd}{2}}{\sin \frac{d}{2}} \sin n \left[\frac{1}{n} \left(\frac{2\pi}{\lambda} e \sin\theta \right) \right]$$

$$R = a \frac{\sin \frac{nd}{2}}{\sin \left[\frac{1}{n} \left(\frac{2\pi}{\lambda} e \sin\theta \right) \right]}$$

$$= a \frac{\sin \frac{\pi e \sin}{\lambda}}{\sin \frac{\pi e \sin}{n\lambda}} = a \frac{\sin \alpha}{\sin \frac{\alpha}{n}} \quad \text{where } \alpha = \frac{\pi e \sin}{\lambda}$$

$$= na \frac{\sin \alpha}{\alpha} = A \frac{\sin \alpha}{\alpha}$$

When $n \rightarrow \infty$ $a \rightarrow 0$ but the product $na = A$ is finite.

$$R = A \frac{\sin \alpha}{\alpha} \Rightarrow \text{Intensity } I = R^2 = A^2 \left[\frac{\sin \alpha}{\alpha} \right]^2$$

Principle maximum:

The value of R will be maximum only when $\alpha = 0$

$$\alpha = \frac{\pi e \sin}{\lambda} = 0 \text{ or } \sin \theta = 0 \text{ or } \theta = 0$$

Minimum intensity position:

The intensity will be minimum when $\sin \alpha = 0$

$$\rightarrow \alpha = \pm \pi, \pm 2\pi, \pm 3\pi, \dots$$

$$\therefore \alpha = \pm m\pi \text{ where } m = 1, 2, 3, \dots$$

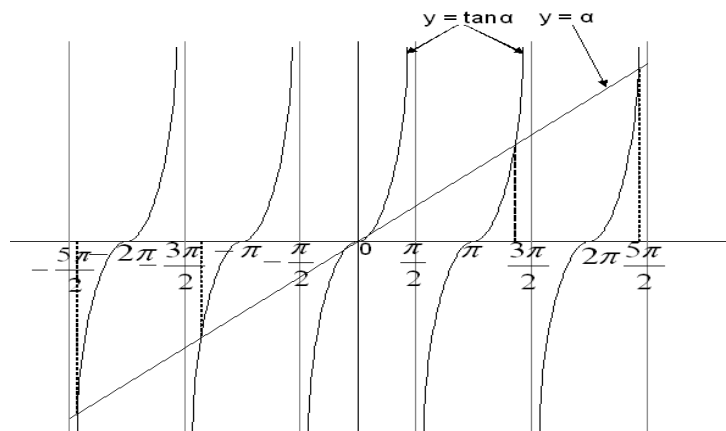
Secondary maxima:

$$\frac{dI}{d\alpha} = \frac{d}{d\alpha} \left[\frac{\sin \alpha}{\alpha} \right]^2 = 0$$

$$A^2 2 \frac{\sin \alpha}{\alpha} \frac{(\alpha \cos \alpha - \sin \alpha)}{\alpha^2} = 0$$

$$\sin \alpha = 0 \text{ or } \alpha \cos \alpha - \sin \alpha = 0$$

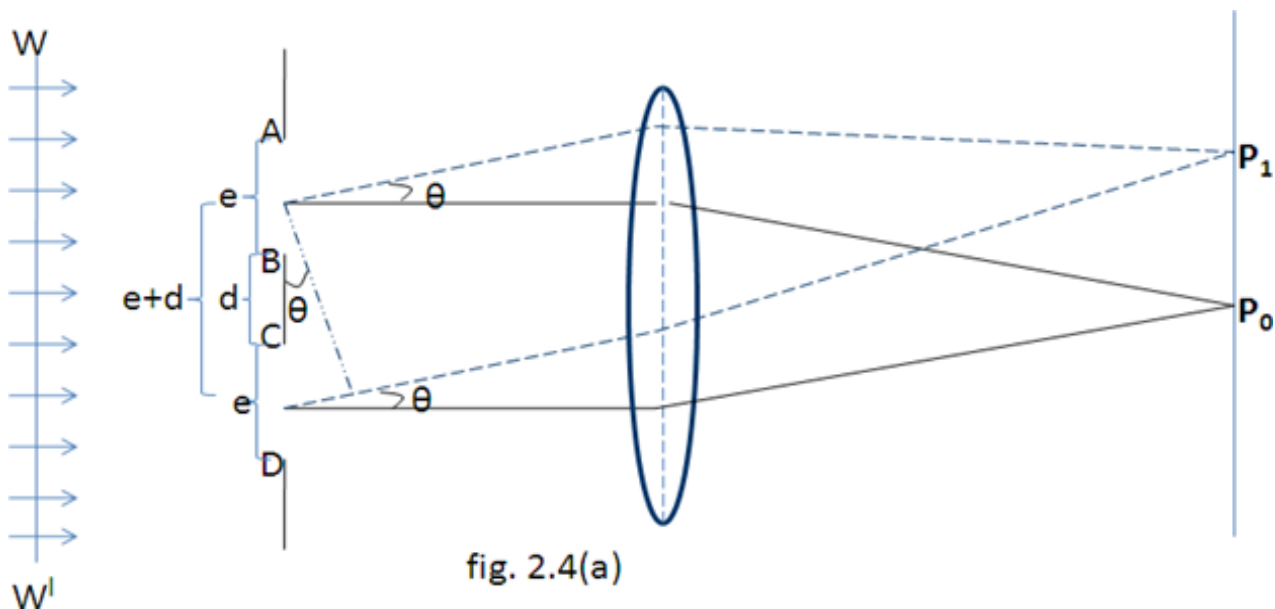
$$\sin \alpha = 0 \text{ or } \alpha = \tan \alpha$$



2. Describe the formation of Fraunhofer's diffraction pattern due double slit:

Let AB and CD be two parallel slits of equal width e and separated by an opaque distance d . Let a parallel beam of monochromatic light of wavelength λ be incident normally upon the two slits. The light diffracted from these slits focused on to the screen XY by means of convex lens. The diffraction at two slits is the combination of diffraction as well as interference.

When the plane wave front is incident normally on both slits, all points within The Slits become the sources of secondary wavelets which travel in all directions.



The secondary waves travelling in the direction of incident light come to focus at p_0 while the secondary wavelets travelling in a direction come to focus at p_1 .

According to the theory of diffraction at a single slit, the resultant amplitude R due to all wavelets diffracted from each slit in a direction θ is given by

$$R = A \frac{\sin \alpha}{\alpha} \quad \text{where } A \text{ is amplitude due to each slit when } \theta=0$$

$$\alpha = \frac{\pi e \sin \theta}{\lambda}$$

The path difference between wavelets from S_1 and S_2 in the direction ' θ '

Triangle $S_1 S_2 K$

$$\sin \theta = S_2 K / S_1 S_2 \quad \text{since } e/2 + d + e/2 = (e+d)$$

$$S_2 K = (e+d) \sin \theta$$

$$\therefore \text{Phase difference } d = \frac{2\pi}{\lambda} (e+d) \sin \theta \quad (1)$$

The resultant amplitude R is given by

$$R = a \frac{\sin \frac{nd}{2}}{\sin \frac{d}{2}} \quad \text{-----1}$$

here

$$a = A \frac{\sin \alpha}{\alpha} \quad \text{from single slit}$$

$n=2$ Number of slits

$$d = \frac{2\pi}{\lambda} (e+d) \sin \theta \quad \text{Phase deference}$$

$$R = A \frac{\sin \alpha}{\alpha} \frac{\sin 2 \left[\frac{2\pi}{\lambda} (e+d) \sin \theta \right]}{2}$$

$$\text{Here } \frac{\pi}{\lambda} (e+d) \sin \theta = \beta$$

$$R = A \frac{\sin \alpha}{\alpha} \frac{\sin 2\beta}{2 \sin \beta \cos \beta}$$

$$R = A \frac{\sin \alpha}{\alpha} \frac{2 \sin \beta \cos \beta}{\sin \beta}$$

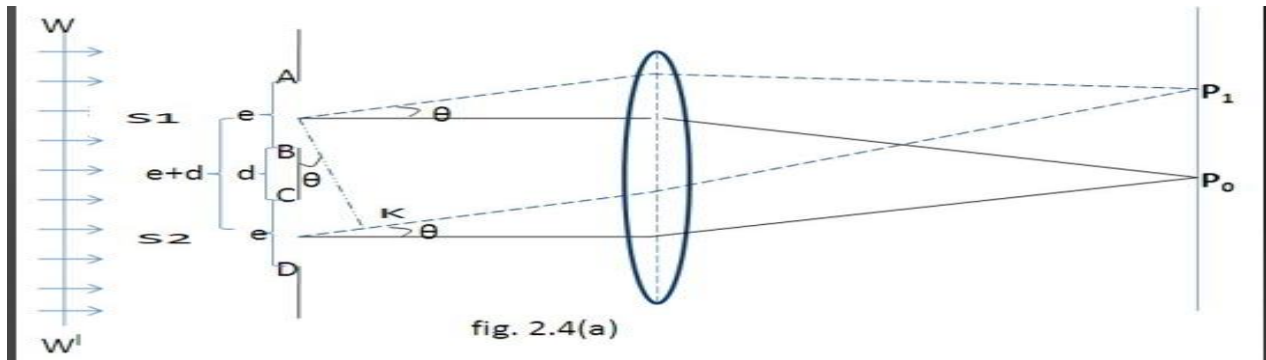
$$R = 2 A \frac{\sin \alpha}{\alpha} \cos \beta \quad \text{since } I = R^2$$

$$I = 4 A^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 \cos^2 \beta$$

$$2 \quad 2 \quad 2$$

3. Describe the formation of Fraunhofer's diffraction pattern due double slit:

an arrangement of 'N' parallel slits on which a plane wave front WW^1 incident normally and diffracted light made to focus on the screen XY by means of convex lens. Let 'e' be the width of each slit and 'd' is the separation between two consecutive slits. The secondary wavelets travelling in the direction of incident light will come to focus at point ' P_0 ' by means of convex lens, that will be a central maximum.



Consider the secondary wavelets travelling in a direction inclined at an angle ' θ ' with the direction of incident light will come to focus at point ' P_1 '. The intensity at point ' P_1 ' may be considered by applying the theory of Fraunhofer diffraction at a single slit. The wavelets proceeding from all points on a slit along the direction ' θ '

are equivalent to a single wave of amplitude $A \frac{\sin \alpha}{\alpha}$ starting from middle point of

the slit (where $\alpha = \frac{\pi e \sin \theta}{\lambda}$).

If there are 'N' slits, then we have 'N' diffracted waves. The path difference between the waves from two consecutive slits is $(e+d) \sin \theta$.

The path difference between wavelets from S_1 and S_2 in the direction ' θ '

Triangle $S_1 S_2 K$

$$\sin \theta = \frac{S_2 K}{S_1 S_2} \quad \text{since} \quad e/2 + d + e/2 = (e+d)$$

$$\text{Path difference } S_2 K = (e+d) \sin \theta$$

$$\therefore \text{Phase difference } \phi = \frac{2\pi}{\lambda} (e+d) \sin \theta$$

Therefore, there is a corresponding phase difference $\frac{2\pi}{\lambda} (e+d) \sin \theta$

By the method of vector addition of amplitudes, the resultant amplitude in the direction ' θ ' will be

$$R = a \frac{\sin \frac{nd}{2}}{\sin \frac{d}{2}} \text{ -----1}$$

$$a = A \frac{\sin \alpha}{\alpha} \text{ from single slit}$$

$n = N$ Number of slits

$$d = \frac{2\pi}{\lambda} (e+d) \sin \theta \text{ Phase difference}$$

$$\sin N \left[\frac{2\pi}{\lambda} (e+d) \sin \theta \right]$$

$$R = A \frac{\sin \alpha}{\alpha} \frac{\sin N \left[\frac{2\pi}{\lambda} (e+d) \sin \theta \right]}{\sin \left[\frac{2\pi}{\lambda} (e+d) \sin \theta \right]} \text{ since } \frac{\pi}{\lambda} (e+d) \sin \theta = \beta$$

$$\frac{\sin \frac{2\pi}{\lambda} (e+d) \sin \theta}{\sin \left[\frac{2\pi}{\lambda} (e+d) \sin \theta \right]}$$

$$R = A \frac{\sin \alpha}{\alpha} \frac{\sin N \beta}{\sin \beta}$$

Principle maxima:

The intensity would be maximum when $\sin \beta = 0 \rightarrow \beta = \pm n\pi$
where $n = 1, 2, 3, \dots$

)

Minima: A series of minima occur, when $\sin N\beta = 0$

$\neq$

Secondary maxima:

$$\frac{dI}{d\beta} = 0$$

$$N \cos N\beta \sin \beta - \sin N\beta \cos \beta = 0 \quad \Rightarrow N \tan \beta = \tan N$$

4. Discuss theory of diffraction grating:

An arrangement consisting of large number of parallel slits of the same width and separated by equal opaque spaces is known as diffraction grating.

Fraunhofer used the first grating consisted of large number of parallel wires placed very closely side by side at regular intervals. The diameter of the wires was of the order of 0.05mm and their spacing, varied from 0.0533mm to 0.687mm. Now gratings are constructed by ruling equidistant parallel lines on a transparent material such as glass, with a fine diamond point.

If 'e' is the width of the slit and 'd' is the width of opaque part between two consequent slits then 'e+d' is called grating element

Formation of Multiple Spectra with Grating:

We know that the principal maxima in a grating are formed in direction θ given by

$$(e+d) \sin \theta = \pm n \lambda$$

Where (e+d) is the grating element, n is the order of the maxima and λ wavelength of the incident light.

Chief characteristics of grating spectra:

1. spectra of different orders are situated symmetrically on both sides of zero order image.
2. Spectral lines are almost straight and quite sharp.
3. Spectral colours are in the order from violet to red.
4. The spectral lines are more and more disappeared as we go to higher orders.
5. Most of the incident intensity goes to zero order and rest is distributed among the other order.

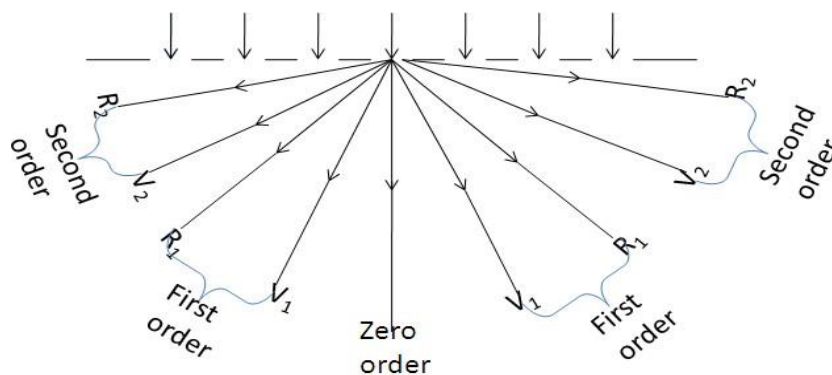


Fig. 2.61a

Maximum number of Orders available with a grating:

The principal maxima in a grating satisfy the condition

$$(e+d) \sin \theta = \pm n \lambda \quad \Rightarrow n = (e+d) \sin \theta / \lambda$$

The maximum angle of deflection is 90° , hence the maxima possible order is

$$n = (e+d) \sin 90^\circ / \lambda$$

$$= (e+d) / \lambda$$

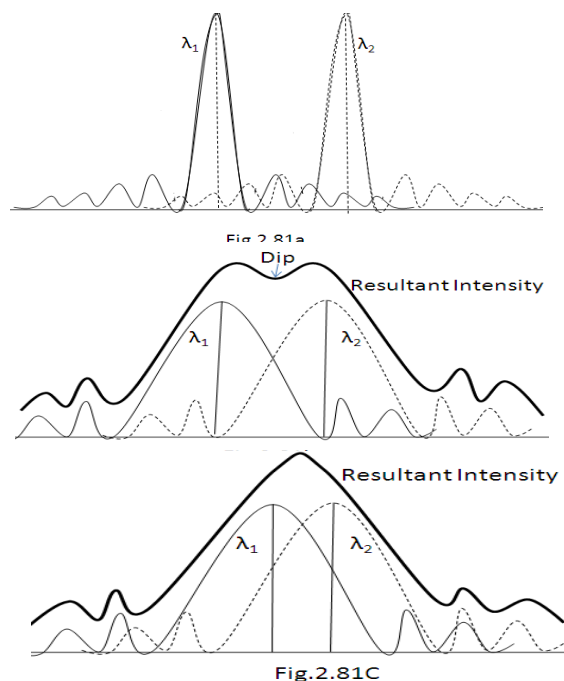
$$= 1/N\lambda \quad \text{since } 1/N = (e+d)$$

5.Explain Rayleigh's Criterion for Resolving power

The ability of the instrument to produce their separate pattern is known as resolving power. The reciprocal of limit of resolution is called the resolving power.

According to Rayleigh criterion, two-point sources are resolvable by an optical instrument when the central maximum in the diffraction pattern of one falls over the first minimum on the diffraction pattern of the other and vice versa.

Let us consider the resolution of two wavelengths ' λ_1 ' and ' λ_2 ' by a grating fig.2.81a shows the intensity curves of the diffraction patterns of two wavelengths. The difference in wavelengths is such that their principal maxima are separately visible. There is a distinct point of zero intensity in between the two. Hence the two wavelengths are well resolved.



Next consider the case shown in fig.2.81b, the difference in wavelengths is smaller and such that the central maximum of one coincides with the first minimum of the other. The dip in the middle of the two central maxima indicating the presence of two different wavelengths. Thus, the two wavelengths can be distinguished from one another they are said to just resolved.

Now consider the case shown in fig.2.81c, the difference in wavelengths is so small that the central maxima corresponding to two wavelengths come still closer, the resultant intensity curve in this case is quite smooth without

any dip. Hence the two wavelengths are not resolved

6. Derive and Explain Resolving power of a grating:

The resolving power of a diffraction grating is defined as the capacity to form separate diffraction maxima of two wavelengths which are very close to each other. This is measured by $\frac{\lambda}{\Delta\lambda}$, where $\Delta\lambda$ is the smallest difference in two wavelengths which are just resolvable by grating and ' λ ' is the mean wavelength.

Let AB represent the surface of a plane transmission grating having grating element $(e+d)$ and N total number of slits (shown in fig 1). Let a beam of light having two wavelengths ' λ ' and ' $\lambda+d\lambda$ ' is normally incident on the grating. XY is the field of view of the telescope, ' P_1 ' is n^{th} primary maximum of a spectral line of wavelength ' λ ' at an angle of diffraction ' θ_n ' and ' P_2 ' is

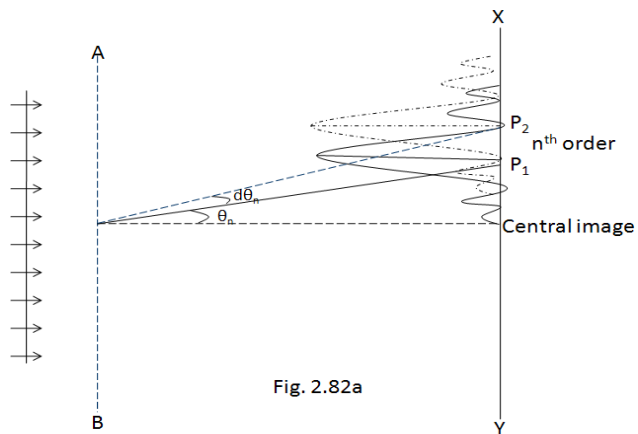


Fig. 2.82a

the n^{th} primary maximum of wavelength $(\lambda+d\lambda)$ at diffraction angle $(\theta_n+d\theta_n)$. According to the Rayleigh criterion, the two wavelengths will be resolved if the position of ' P_2 ' corresponds to the first minimum of ' P_1 ' i.e., the two lines will be resolved if the principal maximum of ' $\lambda+d\lambda$ ' in n^{th} order in a direction $(\theta_n+d\theta_n)$ fall over the first minimum of ' λ ' in the same direction $(\theta_n+d\theta_n)$.

The condition for n^{th} order principal maximum of ' λ ' in the direction ' θ_n ' is given by $(e+d)\sin\theta_n = n\lambda$ (1)

The condition for first minimum adjacent to n^{th} principal maximum in the direction $(\theta_n+d\theta_n)$ is $N(e+d)\sin(\theta_n+d\theta_n) = (nN+1)\lambda$ (2)

The principal maximum of $(\lambda+d\lambda)$ in direction $(\theta_n+d\theta_n)$ is given by

$$(e+d)\sin(\theta_n+d\theta_n) = n(\lambda+d\lambda) \quad (3)$$

Multiplying equation (3) by ' N ' we have

$$N(e+d)\sin(\theta_n+d\theta_n) = N n(\lambda+d\lambda) \quad (4)$$

$$\text{From (2) \& (4) we get } (nN+1)\lambda = nN(\lambda+d\lambda) \quad \Delta\lambda = nN d\lambda \quad \frac{\lambda}{d\lambda} = nN$$

$\therefore$ Resolving power of the grating $= nN$

7. Dispersive power of Grating

The dispersive power of grating is defined as the rate of variation of the angle of diffraction with the wavelength of light

It is noted with $D = d\theta/d\lambda$

We know that condition for principal maxima

$$(e+d) \sin\theta = \pm n \lambda$$

Differentiating the above eqn with respect to λ

$$d\theta/d\lambda = n/(e+d) \cos\theta$$

$$d\theta/d\lambda = nN/\cos\theta \quad \text{since } 1/N = (e+d)$$

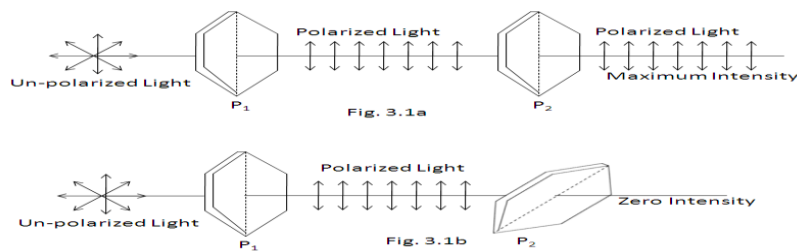
Unit-1

Part-C

POLARIZATION

1.Polarization of light waves:

When ordinary light is passed through a pair of tourmaline crystal plates P_1 and P_2 with their planes at right angles to the direction of propagation of light Fig.3.1(a), the intensity is maximum. But when the plane ' P_2 ' is rotated through 90° Fig.3.1(b), the intensity is minimum. This means when ordinary light is passing through crystals like tourmaline it has acquired the property of one-sidedness, i.e., vibrations confined only to a single line. The phenomenon is called polarization. The light which has acquired the property of one-sidedness is called polarized light.



2.Different types of Polarization

Polarized light can be classified into three types. they are

- 1) Plane polarized light
- 2) Circularly polarized light
- 3) Elliptically polarized light

1)Plane polarized light:

When the amplitude of the electric vector of the light changes and orientation remains constant. The path traced by electric vector is a plane and the light is said to be linearly plane polarized.

2)Circularly polarized light:

When the amplitude of the electric vector of the light remains constant. But the orientation changes then the path traced by electric vector is a circle and the light is said to be circularly polarized light.

3)Elliptically polarized light:

When the amplitude of the electric vector of the light and its orientation are changing the path described by electric vector is an ellipse then the light is said to be Elliptically polarized light

3.Properties of O-ray and E-ray

O-ray refractive index does not vary with direction

The refractive index for E-ray varies with direction

The difference between the refractive index of O-ray and E-ray is called “birefringence”.

The direction in the crystal along which no double refraction takes place is called optic axis.

O-ray and E-ray are polarized at right angles to each other

The vibration direction of O-ray is perpendicular to the plane defined by the direction of propagation and the optic axis.

The vibration direction of E-ray is in the plane as defined above.

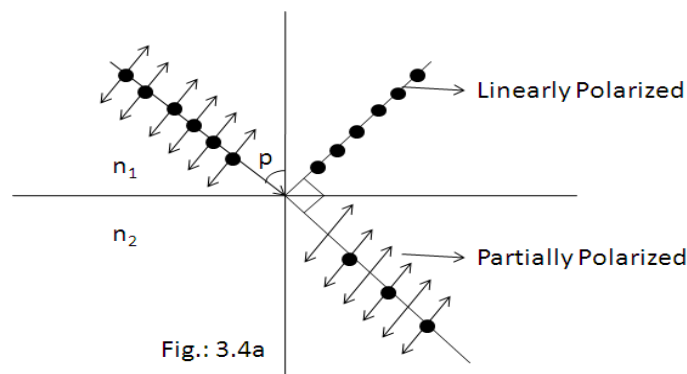
4.Polarization Methods:

The methods by which the polarized waves are produced are

- Polarization by reflection.
- Polarization by double refraction.
- Polarization by selective absorption.
- Polarization by scattering.

1.Polarization by reflection:

When natural light strikes a reflecting surface, there is found to be a preferential reflection for those waves in which the electric vector is vibrating perpendicular to the plane of incidence. At one particular angle of incidence known as polarizing angle(Brewster angle) only the light for which the electric vector is perpendicular to the plane of incidence is reflected. The components perpendicular to the plane of incidence about 15% are reflected if the reflecting surface is transparent material like glass. Hence the reflecting light is weak and completely linearly polarized.



2.Polarization by double refraction:

When the unpolarized beam enters an anisotropic crystal, it splits up into two beams, each of them being characterized by certain state of polarization. One beam is found to obey Snell's law and is called O-ray (ordinary ray), the other beam does not obey Snell's law and is not even found to be in the plane of incidence; it is called E-ray (extraordinary ray). This phenomenon is called double refraction.

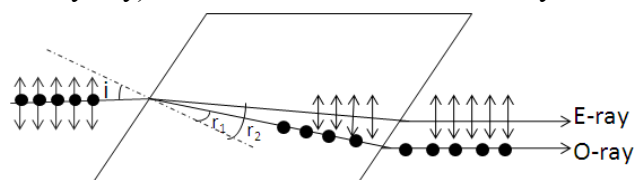


Fig.:3.5a Double refraction

3. Polarization by a selective Absorption:

There are certain crystals and minerals of double refracting class which have the property of absorbing one of the double refracting beams to an extent more than the other. The crystal showing the property is termed as dichroic and the phenomenon is known as dichroism. The phenomenon of selective absorption is shown by number of substances but most notable is tourmaline.

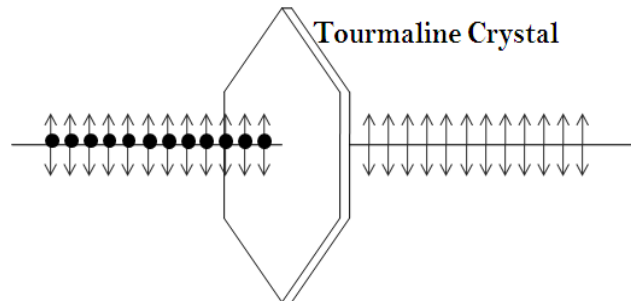


Fig. 3.7a: Polarization by selective absorption

4. Polarization by scattering:

When a narrow beam of natural light is incident on a transparent medium containing a suspension of ultra microscopic particles the light is Scattered and partially polarized. the degree of polarization depends on the angle of scattering. the beam scattered at 90 degrees with respect to the incident direction is a plane polarized therefore the intensity of scattering light is proportional to the reciprocal of fourth power of wavelength of light this is called Skittering of polarization

5. CONSTRUCTION AND WORKING OF NICOL'S PRISM:

NICOL'S PRISM is derived for producing plane polarized light by eliminating O-ray by utilizing the phenomenon of total internal reflection at thin film of Canada balsam separating two pieces of calcite. This is also used for analyzing polarized light.

CONSTRUCTION: A calcite crystal whose length is three times as its width is taken. The end faces of crystal are grinded in such a way that the angles in the principal section become 68° and 112° instead of 71° and 109° (shown in Fig.3.6). This is to increase the field of view. The crystal is cut along a diagonal and cemented back together again with Canada Balsam. The refractive index of Canada balsam lies between the refractive indices of the O-ray and E-ray for calcite.

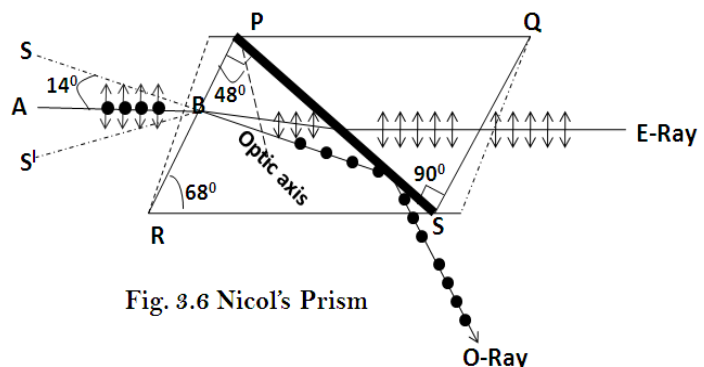


Fig. 3.6 Nicol's Prism

ACTION:

When a beam of light AB enters the face PR in direction parallel to the long side, it is doubly refracted into ordinary plane polarized beam BO and extra-ordinary plane polarized beam BE. Canada balsam acts as a rarer medium for O-ray and denser medium for E-ray and the dimensions of the crystal are so chosen that the angle of incidence of ordinary ray at the calcite balsam surface becomes greater than the corresponding crystal angle. Under these conditions the O-ray is completely reflected at calcite balsam surface and the E-ray is transmitted with no appreciable loss in intensity. The E-ray is plane polarized having vibrations parallel to principal plane hence the light emerging from the Nicol's prism is plane polarized.

Uses: Nicol's prism can be used both as polarizer and an analyzer.

6. Explain Quarter wave plate and Half wave plate:

Quarter wave plate:

If the thickness of the crystal plate is such that it introduces a phase difference of $\frac{\pi}{2}$ radians or a path difference of $\frac{\lambda}{4}$ then it is called a quarter wave plate.

To calculate thickness of such a plate:

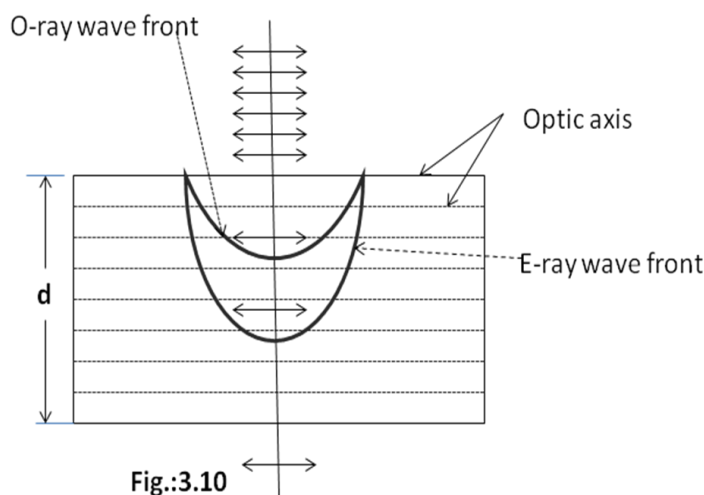
$$\text{Phase difference} = 2\pi \frac{d}{\lambda} (\mu_o - \mu_e) = \frac{\pi}{2} \text{ for quarter wave}$$

$$d_{\text{Quarter wave plate}} = \frac{\lambda}{4(\mu_o - \mu_e)}$$

Half wave plate: If the thickness of a calcite crystal plate, cut with its faces parallel to optic axis, is such that it introduces phase difference of π or a path difference of $\frac{\lambda}{2}$ between O-ray and E-ray waves then it is called a half wave plate.

$$\text{Phase difference} = 2\pi \frac{d}{\lambda} (\mu_o - \mu_e) = \pi$$

$$d_{\text{Half Wave plate}} = \frac{\lambda}{2(\mu_o - \mu_e)}$$



Describe the

UNIT-II. PART-B.

①

BRAGG'S X-RAY SPECTROMETER

The experimental arrangement of Bragg's spectrometer as shown in below fig.

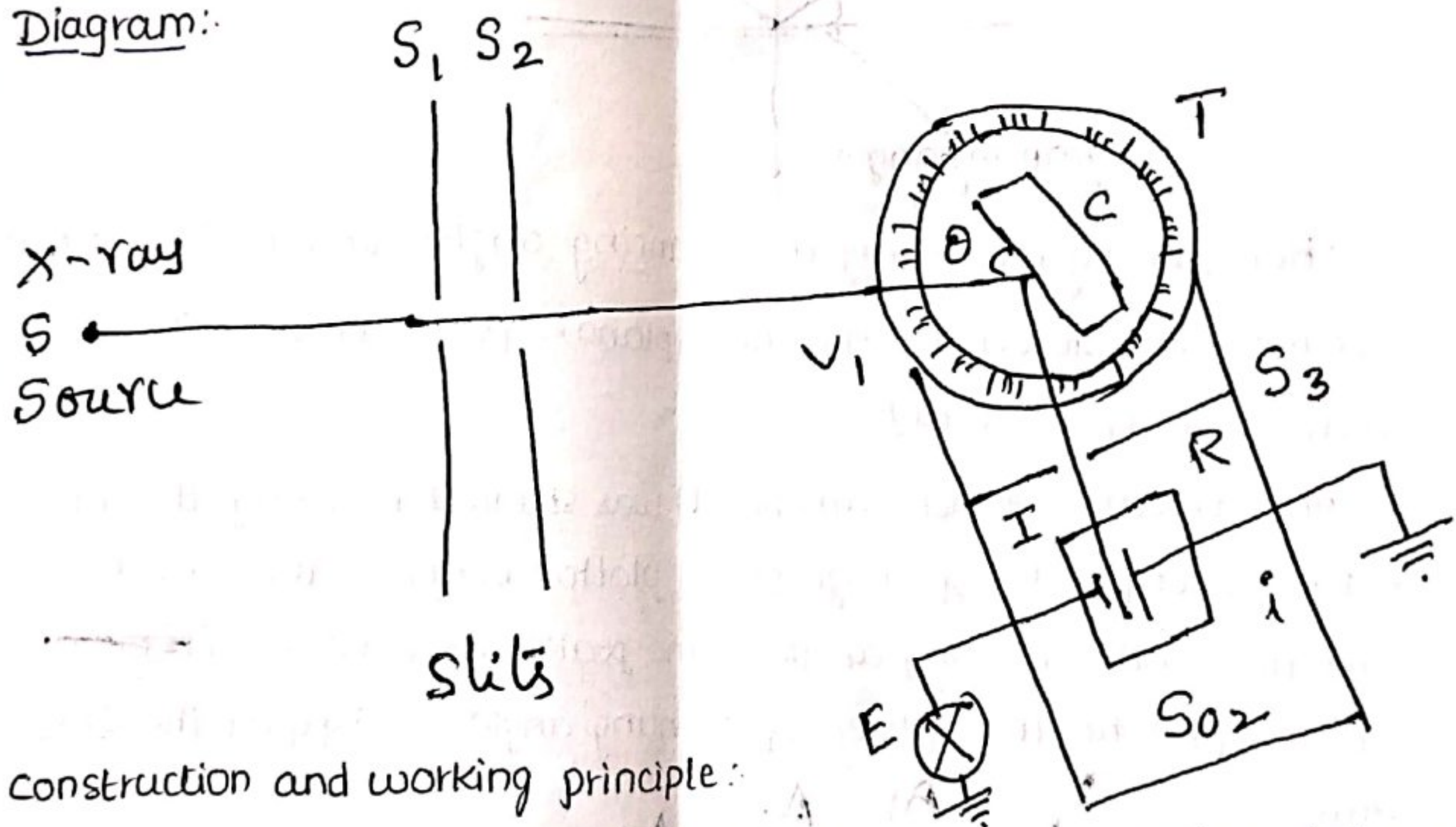
Bragg's x-ray spectrometer device is designed to study the structural properties of crystal.

Main parts :

Bragg's spectrometer consists of three main parts They are

1. source of x-ray
2. crystal Grating
3. Detector consisting of an ionization chamber

Diagram:



Construction and working principle:

Bragg designed a device to study the structural properties of the crystals by diffracting x-rays from the crystals and is known as Bragg's x-ray spectrometer. The experimental arrangement of Bragg's spectrometer as shown in Fig. 10.17. The x-rays from an x-ray tube are allowed to pass through slits S_1 and S_2 to become a fine narrow beam and allowed to incident on a crystal which is placed at the centre of the turntable. The table is mounted on a graduated scale. Bragg used an ionization chamber as a detector to measure the intensities of the x-rays which are scattered from the crystal. The crystal is

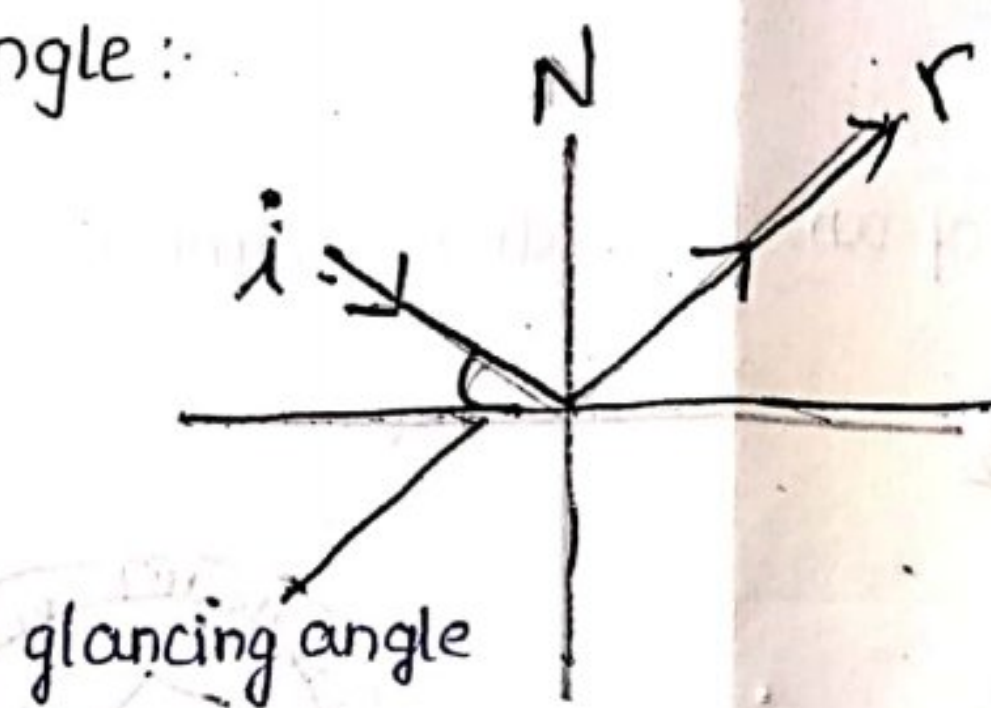
rotated by means of the turntable so as to increase the glancing angle at which x-rays are incident at the exposed face of the crystal. The angles, for which the reflection intensities becomes maximum, gives the values of θ of equation

$$2d \sin \theta = n\lambda, \text{ where, } n \text{ is order of the spectrum}$$

If θ_1, θ_2 and θ_3 are the glancing angles of first, second and third order reflections respectively then,

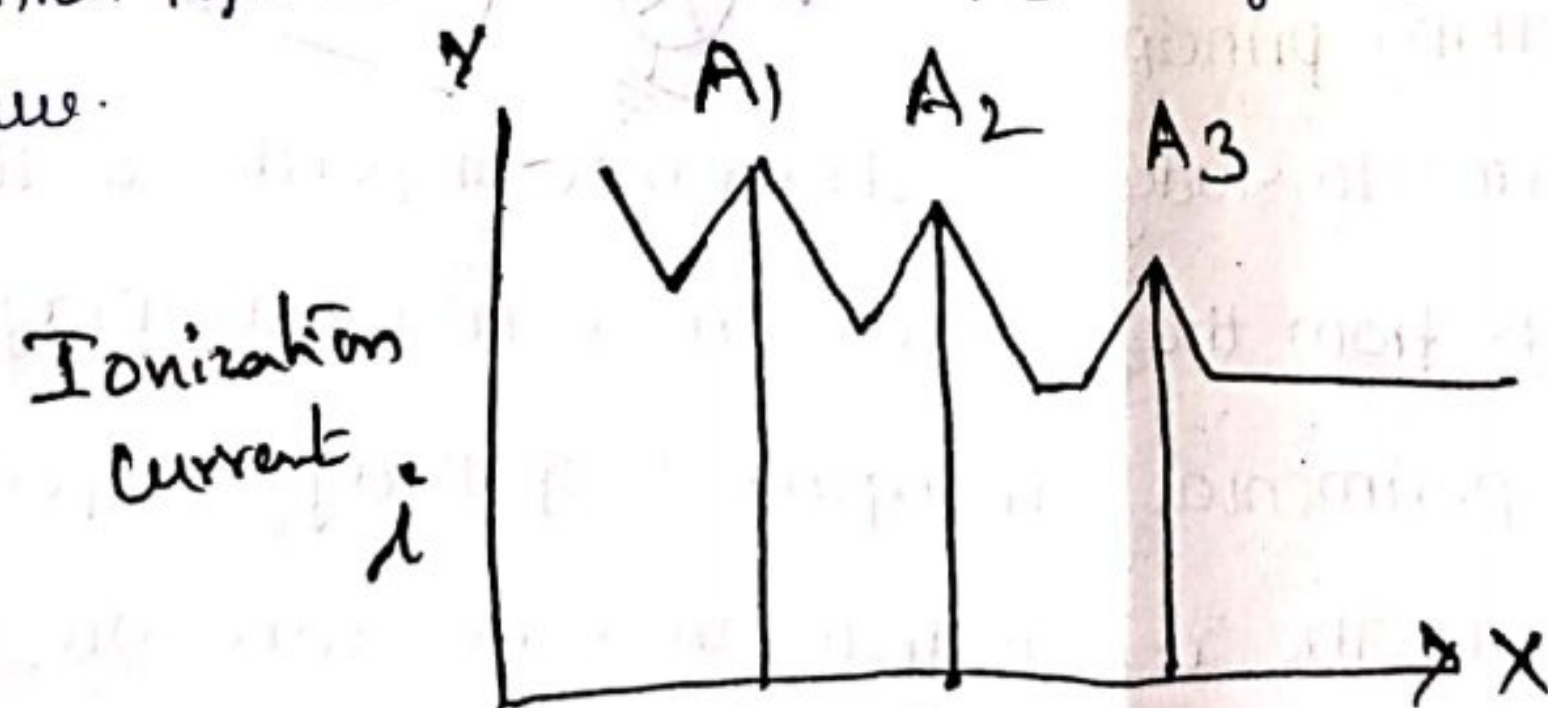
$$\sin \theta_1 : \sin \theta_2 : \sin \theta_3 = 1 : 2 : 3$$

Glancing Angle:



Therefore, by measuring the glancing angles at which reflections occur, we can determine the inter-planar spacing knowing the wavelength of the x-rays.

The structure of rock-salt (NaCl) was studied by using the Bragg's x-ray spectrometer. If a graph is plotted between the ionization current and the glancing angle, some peaks are obtained (Fig. 18.18) which represent the values of glancing angles satisfying the Bragg's law.



A_1, A_2, A_3 are set of $\theta_1, \theta_2, \theta_3$ maximum intensity corresponding, $\theta_1, \theta_2, \theta_3$.
Application:

1. we can determined wavelength (λ) of source.
2. we can find out lattice spacing (d).
3. we can analyzed crystal structure.
4. To calculate crystallite size.

LAUE METHOD OF X-RAY DIFFRACTION

(2)

In 1912 Max von Laue proposed that crystals can be used as three dimensional grating to diffract x-ray.

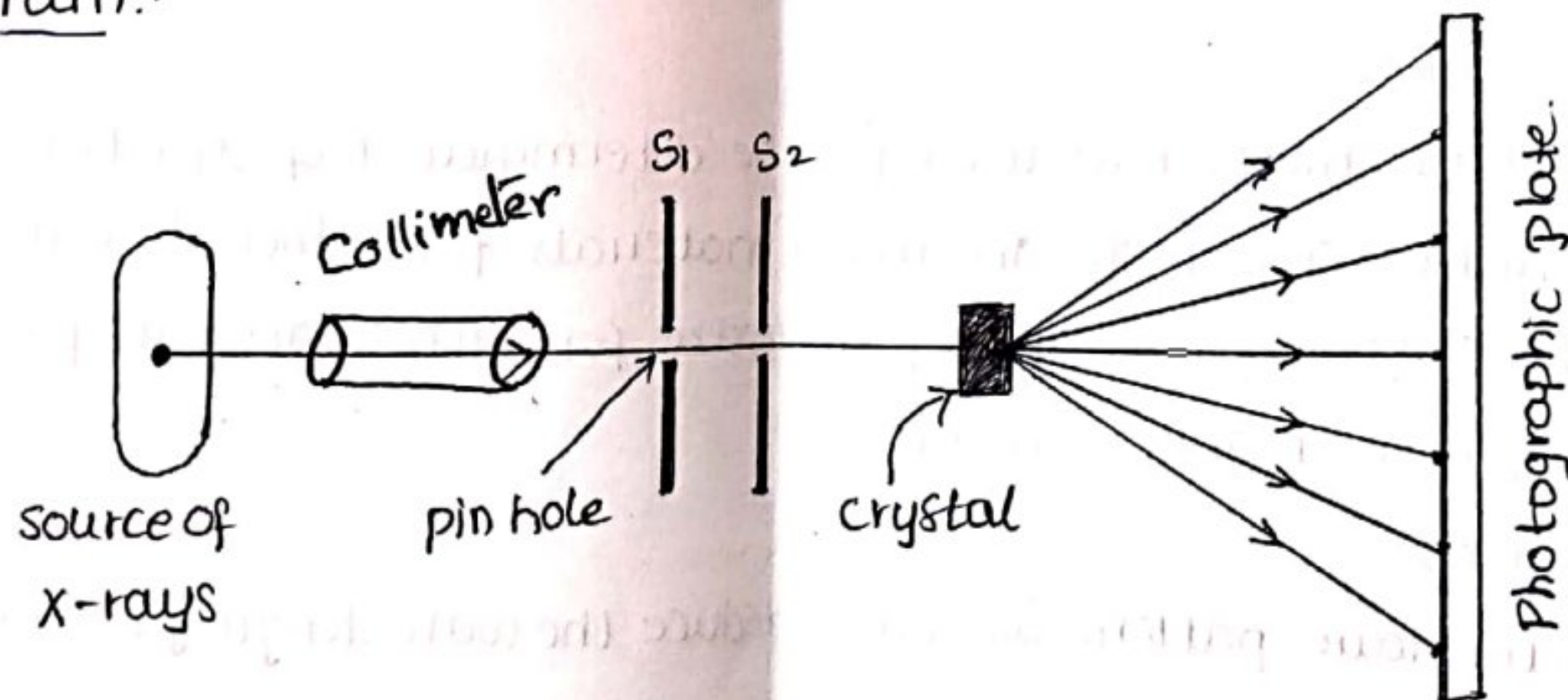
The Laue method is used for determination of the structure of a single crystal. The experimental arrangement consists of source of x-rays, pinhole collimator, crystal holder and photographic film or fluorescent screen as shown in Fig. below diagram.

Main parts:

1. source of light
2. collimeter
3. Sample crystal (ZnS)
4. A photographic plate (or) fluorescent screen

The experimental arrangement for a transmission Laue pattern is illustrated as shown in below diagram, the collimated continuous spectrum passes through thin slit that is collimeter. The diffracted beams are registered on photographic film.

Diagram:

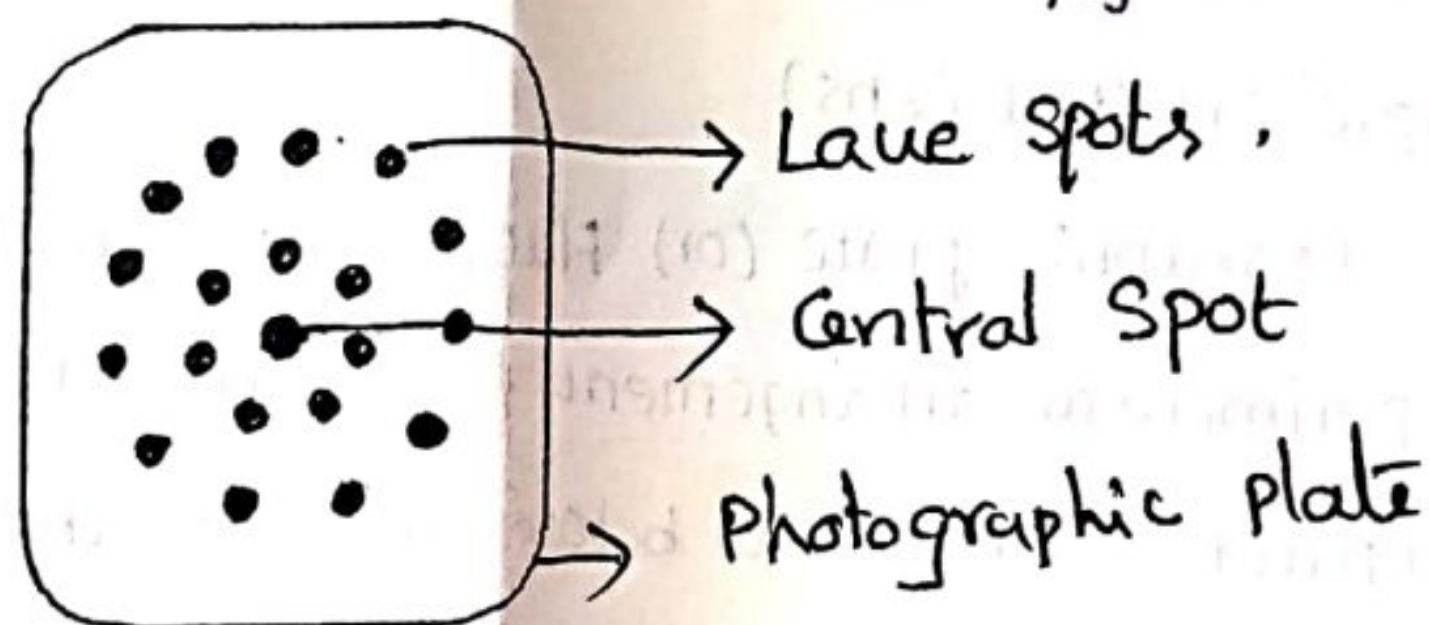


Construction and working principle:

The source produces x-rays of short wavelength (0.2 to $2\text{\AA}$) which are allowed to pass through a slit system (S_1 and S_2) to become a fine and sharp beam. This beam is allowed to be incident on a crystal, placed in a crystal holder. After passing through the crystal, the x-rays are diffracted and recorded on a photographic plate, because

the X-rays penetrate into the crystal and are scattered from different atomic diffraction centers. This is possible because there is a whole range of wavelengths in the continuous spectrum and there will be discrete values of ' λ ' satisfying the Bragg's condition, $2d \sin \theta = n\lambda$.

The diffraction pattern consists of a central spot surrounded by concentric circular spots. The symmetrical pattern of spots is known as Laue pattern. Each spot in the Laue pattern corresponds to an interference maximum for a set of crystal planes satisfying the Bragg's condition, for a particular wavelength scattered from the incident beam. The atomic arrangement can be understood from a study of the position and intensities of the Laue spots shown in below figure.



From this method it is possible to find the lattice constant and the interplanar spacing. The Laue method is useful for the determination of crystal orientation, symmetry and also to study the crystalline imperfections.

The Laue method is used for the determination of structure of single crystals only. There are many materials for which it is impossible to obtain single crystals of required size. For such materials powder photography is highly suitable.

Applications:

1. Using this Laue pattern we can deduce the wave length present in incident beam.
2. If we know the values of wave length present 3 we can find the atomic arrangement in the crystal.
3. which is helps to determine the shape of the unit cell but not determined the structure of material.
4. It is also helps to understand the defects present in the crystal.
5. The Laue method is mainly used to determine orientation of large single crystal.

Explain The Principle, and Procedure, of , ③ Debye - Scherrer Method (powder Method) :

The experimental arrangement of powder diffraction method is shown in below fig.

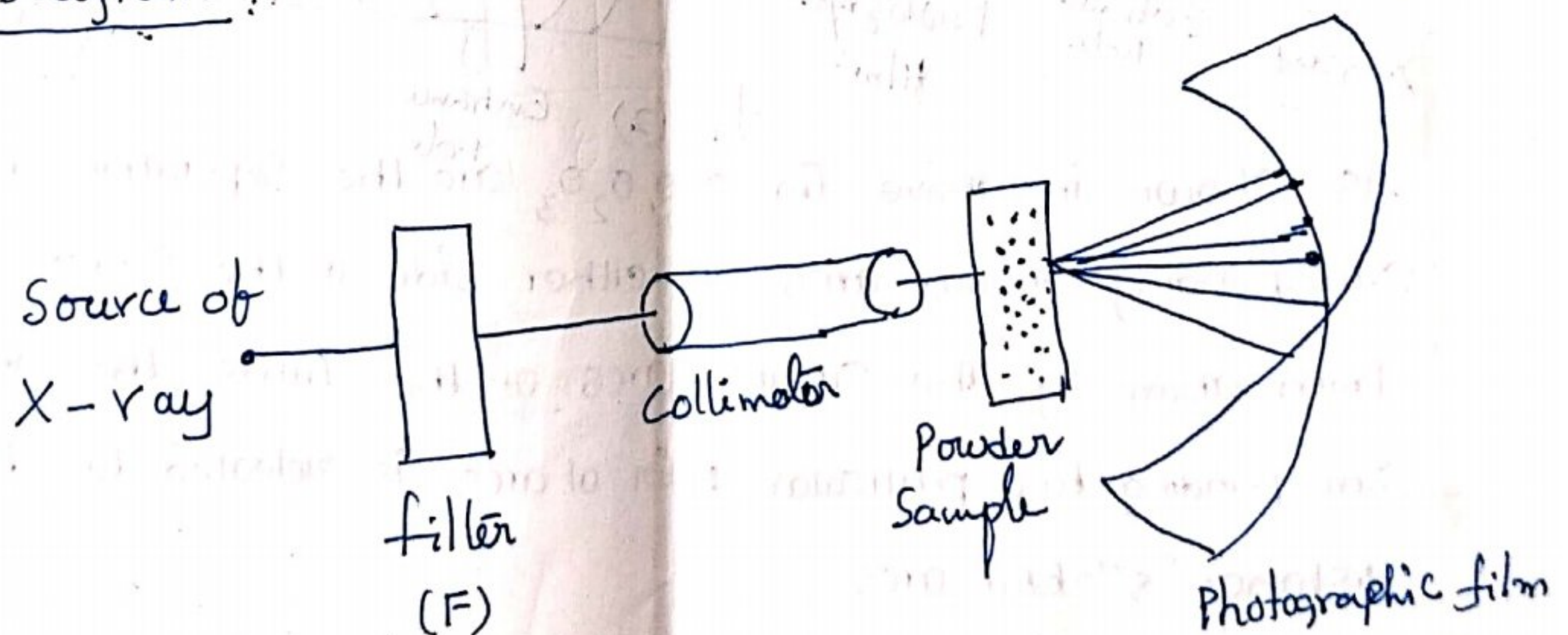
It is used to study tiny crystallite in the form of powder.

Main parts of the Experiment :

- (1) Source of the X-ray
- (2) Filter
- (3) Collimator
- (4) Crystal powder Sample
- (5) photographic plate

Bragg's Spectrometer is useful only in the case of single crystal. Debye Scherrer and Hill developed the powder diffraction method to study the structure of the crystals. In this method, the sample is used in the form of fine powder containing a large number of tiny crystallites with random orientation.

Diagram :



Fig(1)

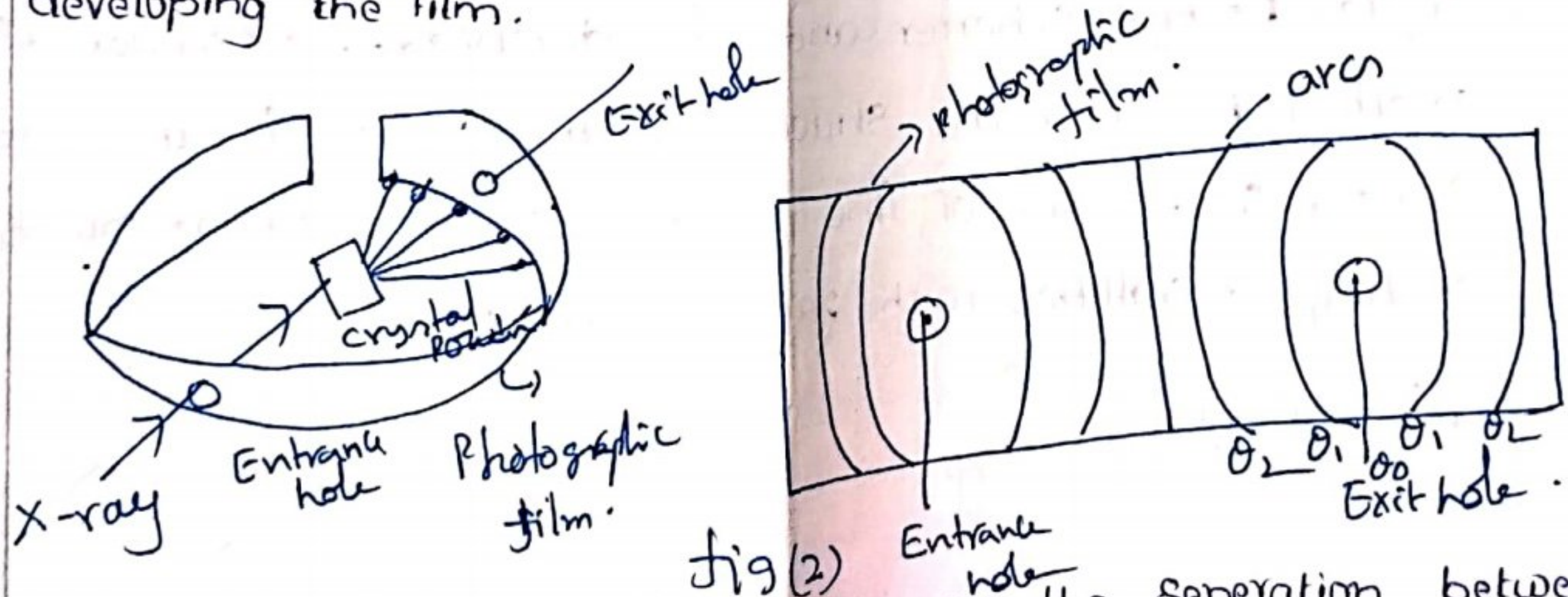
The powder sample is filled in a thin Capillary tube "c".
The Capillary tube "c" should be Mon. diffracting materials.

Construction and working principle:

A Monochromatic beam of x-ray coming out of filter and fall on the specimen powder in c.

The powder consist of large number of very small Crystallite with random orientation.

As a result almost all the possible values of θ and d are available in The bragg's law $2d \sin \theta = n\lambda$ with diffracting angle " θ ". Different lines will be obtained for different combination of n for particular value of " d ". In the experiment. The scattered x-ray are Incident on photographic film and p. This pattern we get recorded on photographic film and exhibited after developing the film.



As shown in above fig $\theta_0, \theta_1, \theta_2, \theta_3$ are the separation between corresponding circular arcs on either side of the center.

From above fig The circular arcs on the films the angle " θ " corresponding to a particular pair of arcs is related to the distance " s " b/w arcs

And we get " θ " from the Experiment

(4)

$$4\theta = \frac{S}{R} \text{ radians. or } \frac{57.296 S}{R}$$

Where S is distance b/w the two diffraction lines related to the Bragg's angle

R is radius of Camera

Applications

- (1) Using these θ values in the Bragg's law we can calculate the Lattice Spacing " d "
- (2) This Method provides information about
 - Size of the Crystal
 - Presence of Impurities
 - distortions in the Crystal
 - Preferred orientation of the Crystal
- (3) Using this Method determine types of Cubic Structures
- (4) determine value for its lattice parameters

— 0 —

4) State and Explain (Bragg's X-ray Equation)

Diffraction of x-ray was first discovered by Laue in 1912. According to him, crystals are just as three dimensional grating.

In 1913, Bragg's studied the diffraction of x-ray from the various (planes) parallel planes of crystal and obtain condition are called Bragg's Condition.

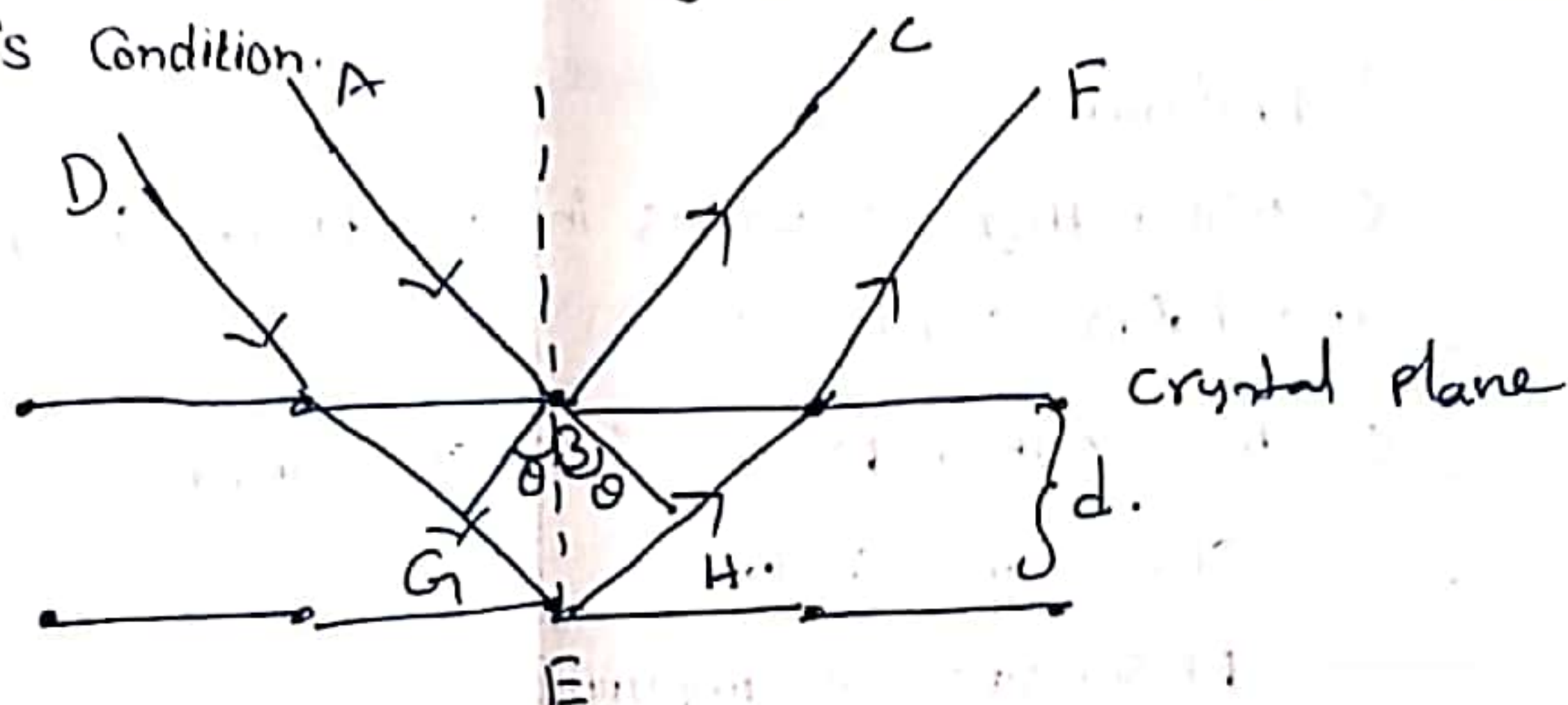


fig (1)

x-rays are electromagnetic waves of very shortest wave length ($\lambda = 1\text{\AA}$) and they exhibit wave phenomena like interference and diffraction: ordinary optical device like narrow slits diffraction don't produce with x-rays but crystal lattice acts as three dimensional grating. Therefore, the crystals are produced diffraction with x-rays.

From the above figure,

AB and DE are monochromatic rays, A is reflected at B and travel along C similarly D is refracted at E and along F and 'd' is inter planar distance of two crystal planes and then BG, BH are normals drawn to the DE and EF from B.

Suppose, a monochromatic light be of wavelength ' λ ' be incident on crystal grating as shown in above figure the two rays ABC, DEF travel different distance.

So, they have some path difference (ie)

(5)

$$GE + EH$$

To calculate these path difference,
we consider two Δ 's from diagram

From ΔBGE

$$\sin \theta = \frac{GE}{BE} \quad (BE = d)$$

$$\frac{GE}{d} = \sin \theta$$

$$\boxed{GE = d \sin \theta} \quad - (1)$$

From ΔBHE ,

$$\sin \theta = \frac{EH}{BE}$$

$$\frac{EH}{d} = \sin \theta$$

$$\boxed{EH = d \sin \theta} \quad - (2)$$

$$\begin{aligned} \text{From eqn (1), (2)} \quad GE + EH &= d \sin \theta + d \sin \theta \\ &= 2d \sin \theta \end{aligned}$$

$$\text{Now, path difference} = 2d \sin \theta$$

$\therefore$ The path difference must be equal to $n\lambda$
(as to get bright fringes - Condition)

$$\boxed{n\lambda = 2d \sin \theta}$$

This is called Bragg's law

n - order of spectrum

λ - wave length

d - inter planar distance

θ - Angle of diffraction

Applications:

- 1) Bragg's law is useful for measuring wave length of light
- 2) Determine the lattice spacing of the crystal
- 3) Identification of Crystalline material.
- 4) determine lattice parameter.

— o —

M 2.80



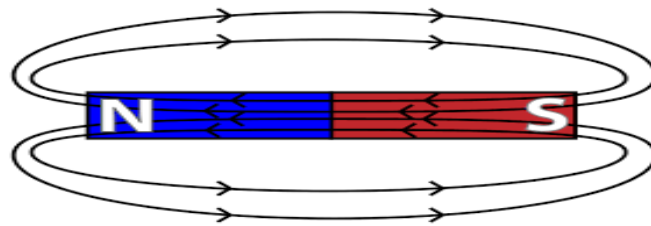
Unit-3

Part-A

BASIC DEFINITIONS

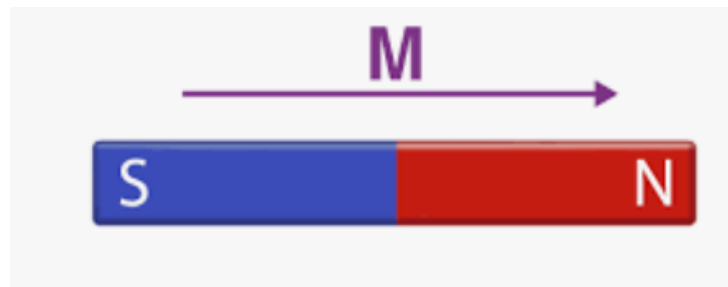
Magnetic dipoles

An arrangement of two magnetic poles of equal and opposite strengths separated by a finite distance is called a magnetic dipole. Two poles of a magnetic dipole or a magnet are of equal strength and opposite nature. The line joining the poles of the magnet is called the magnetic axis.



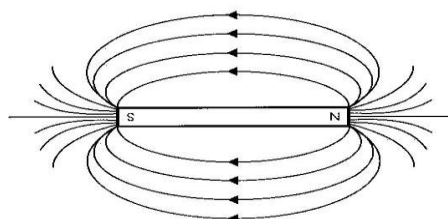
Magnetic dipole moment

The magnetic moment defined as the product of its strength (m) and distance between the two poles ($2l$) is called the magnetic moment



Magnetic field

The magnetic effects are generally exhibited at two ends of a bar magnet. these two ends are called magnetic poles. that is the north and South poles. it is found that the magnetic poles always in pair and cannot be isolated the space around a bar magnet where it is influence is experienced is known as the magnetic field



Magnetic induction or magnetic flux density(B)

It is defined as the number of magnetic lines of force passing perpendicularly through one unit area due to both applied field and magnetization field is called magnetic induction or magnetic flux density

it is noted with $B = \mu \times H$ Tesls

here B is magnetic induction

μ is permeability

H is magnetic field intensity

Magnetic field intensity or magnetic field strength (H)

It is defined as the force experienced by one unit of a north pole placed at a point in the field

it is noted with $H = \frac{B}{\mu}$ ampere/m

here H is magnetic field intensity

B is magnetic induction

μ is permeability

Permeability (μ)

The Permeability is defined as the ratio between magnetic induction and magnetic field intensity at a given point in that medium is called permeability

it is noted with $\mu = \frac{B}{H}$ Henry /m

here μ is permeability

H is magnetic field intensity

B is magnetic induction,

Relative permeability

it is defined as the ratio permeability of medium and permeability of free space is called relative permeability

it is noted with $\mu_r = \frac{\mu}{\mu_0}$ No units

here μ_r is relative permeability

μ is permeability of medium

μ_0 is permeability of free space

The intensity of magnetization (M)

It is defined as the magnetic substance place in the magnetic field, it gets some magnetic dipole moment therefore the magnetic dipole moment for 1 unit volume is call intensity of magnetization

it is noted with $M = \frac{p}{v}$ ampere/m

here **M** is intensity of magnetization

p is Magnetic dipoles moment

v is unit volume of material

Magnetic susceptibility (χ)

It is defined as the ratio of the intensity of magnetization and the magnetic field intensity is called magnetic susceptibility

it is noted with $\chi = \frac{M}{H}$ **No unit**

here χ is magnetic susceptibility, M is intensity of magnetization, H is Magnetic field intensity

Relation between B, H, and M

The good magnetic materials should have high permeability and high susceptibility

When a magnetic substance of permeability placed in a magnetic field then the magnetic flux density is

$$\mathbf{B} = \mu \times \mathbf{H}$$

here flux density has two components one is flux due to applied field and another one is due to magnetization

therefore, resultant flux is some of these two.

the flux density only due to apply field is $\mu_0 H$

similarly due to the magnetization is $\mu_0 M$

so, the resultant flux density is $\mathbf{B} = (\mu_0 H + \mu_0 M)$

$$\mathbf{B} = \mu_0(H + M) \text{ -----1}$$

$$\text{We know that } \mathbf{B} = \mu \times \mathbf{H} \text{ -----2}$$

$$\text{Eqe-2 in 1} \quad \mu \times \mathbf{H} = \mu_0(H + M)$$

$$\frac{\mu}{\mu_0} = \frac{H+M}{H}$$

$$\frac{\mu}{\mu_0} = \frac{1+M}{H}$$

$$\text{since } \frac{M}{H} = \chi$$

$$\frac{\mu}{\mu_0} = \mu_r$$

Therefore $\mu_r = (1 + \chi)$

Origin of the magnetic moment (Bohr magneton)

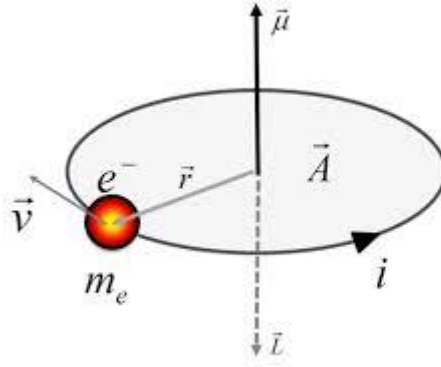
The magnetic moment contributed by a single atom is known as Bohr magneton. mathematically it can be represented as

$$\mu_B = \frac{eh}{4\pi m}$$

Except diamagnetic materials in all magnetic materials of atoms possess a permanent magnetic moment even though the absence of external field, because of the spin motion of the electron and spin motion of nucleus.

we know that electrons are moving around the nucleus in elliptical orbits so electron Orbit can be considered as a close current loop.

if I is a current passing through the closed current loop and A is the area of the close current loop then the magnetic moment due to this close the current loop is given by



$$\vec{\mu} = i \times \vec{A} \text{ -----(1)}$$

$$A \text{ is } \pi r^2 \text{ -----(2)}$$

because the loop is considered as circle

If T is the time taken by electron for the completion of one Revolution, then the current is flowing through the closed

$$\text{loop is } i = \frac{e}{T} \text{ -----(3)}$$

e is charge of electron

T is time taken for one Revolution

eqe 2 and 3 in 1 and we get

$$\text{Therefore } \vec{\mu} = \frac{e}{T} \times \pi r^2 \text{ -----(4)}$$

$$\text{we know that angular velocity } \omega = \frac{2\pi}{T} \text{ Then } T = \frac{2\pi}{\omega} \text{ -----(5)}$$

eqe 5 in 4 we get

$$\mu = \frac{e}{2\pi} \times \pi r^2 \omega$$

$$= \frac{e r^2 \omega}{2} \text{-----(6)}$$

we know that the angular momentum associated with moving particle

Therefore $L = mvr$

here m is mass of the electron

v is linear velocity $= r \omega$

r is radius of loop

Therefore $L = m r^2 \omega \text{-----(7)}$

compare eqe 6 and 7 and we get

$$\mu = \frac{e}{2m} \times L \quad \text{since } L = l \hbar = \frac{lh}{2\pi} \text{ since } \hbar = \frac{h}{2\pi}$$

$$\mu = \frac{e}{2m} \times \frac{lh}{2\pi}$$

$$\mu = \frac{eh}{4\pi m} \times l \text{-----(8)}$$

if $l = 0$ $\mu = 0$

the quantity of $\frac{eh}{4\pi m}$ represent a small unit of magnetic moment, this is called **Bhor magnetic moment**

$$\mu = \frac{eh}{4\pi m} = 9.27 \times 10^{-34} \text{ ampere/m}^2$$

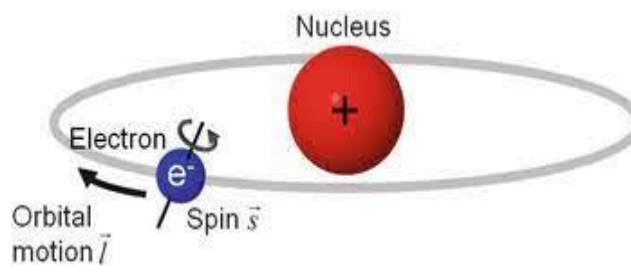
Electron spin motion

Besides the orbital motion of an electron, are it rotates about its own axis. magnetic moment associated with a spinning of the electron is called spin magnetic moment. and the electronic spin dipole moment is given by

$$\mu = - \left(\frac{e}{2m} \right) S \quad \text{and} \quad S = \sqrt{s(s+1)}$$

where S is the spin angular momentum and is define as the ratio of the magnetic dipole moment of an electron due to its spin motion and the spin angular momentum

$$\mu = 9.4 \times 10^{-24} \text{ ampere/m}^2$$



Nuclear spin motion

The atomic nucleus processes intrinsic Spin is called nuclear spin and hence a magnetic moment is associated with this. As the electron magnetic moment is expressed in the unit of Bohr magnetism. the nuclear magnetic moment is expressed in the unit of nuclear magneton μ_n

$$\mu_n = \frac{eh}{4\pi M} = 5.05 \times 10^{-24} \text{ ampere/m}^2$$

where M mass of the proton

Classification of magnetic materials

Based on magnetic dipole moment the magnetic materials classified under two categories

- 1.zero magnetic materials
- 2.permanent magnetic material

zero magnetic materials are Diamagnetic materials

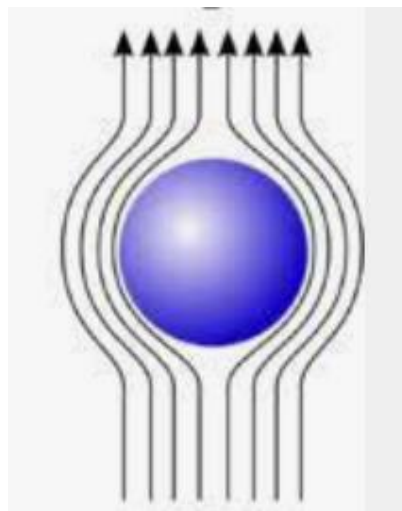
the permanent magnetic material is again classified into four categories

- 1.Para magnetic material
- 2.Ferromagnetic material
- 3.Anti-ferromagnetic material
- 4.Ferrimagnetic material(ferrites)

Diamagnetic materials

if in material the arrangement of orbits and the orientation is such that the vector sum of magnetic moment is zero then the material is known as diamagnetic material.

the diamagnetic materials do not possess a permanent magnetic moment, when the material is placed in an external magnetic field the orbits of the electrons undergo a precessional motion, this is called Larmor's motion.



Spin alignment of diamagnetic materials

Properties of diamagnetic materials

Diamagnetic materials cannot be magnetized

The lines of force do not pass through the specimen

The magnetic susceptibility is negative

Diamagnetic materials, not effect by temperature

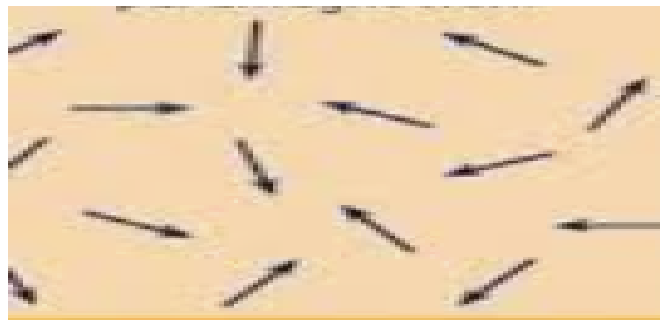
Relative permeability is less than 1

Examples -bismuth, zine, gold, silver, copper, diamond, and water

Paramagnetic material

If in material the arrangement of orbits and orientation is such that the vector sum of magnetic moment is non zero this type of material for call paramagnetic materials.

the paramagnetic materials whose molecules possess a net permanent magnetic moment even in the absence of an external magnetic field this magnetic moment is randomly oriented and hence the net magnetization of the material is zero. when a field is applied to the material that magnetic dipoles is aligned in the direction of the external field so that it becomes easily magnetized

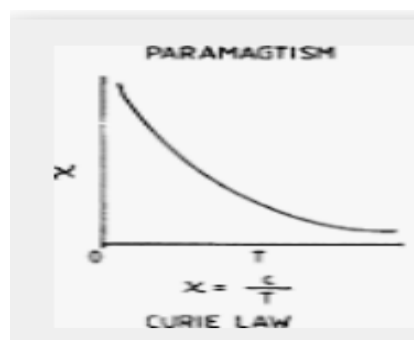


Spin alignment of paramagnetic materials

Properties of paramagnetic materials

Paramagnetic materials possess a permanent dipole moment
susceptibility is positive and small

The susceptibility is inversely proportional to the temperature $\chi = \frac{C}{T}$



The Spin magnetic moment are randomly oriented

The presence of the magnetic field, magnetic lines of force are attracted towards the center of the material

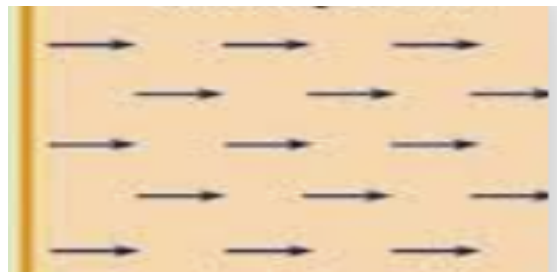
The relative permeability is greater than one but small

Examples -aluminum, Platinum

Ferro magnetic material

If in material the arrangement of orbits so orientation is such that the atoms have a hole possess a large number of magnetic moments, this type of material are called ferro magnetic materials

When a ferromagnetic substance placed in the external magnetic field, they easily magnetized the spin magnetic moments of atoms become align parallel to one other even in the absence of an external magnetic field and possess magnetization this is called spontaneous magnetization



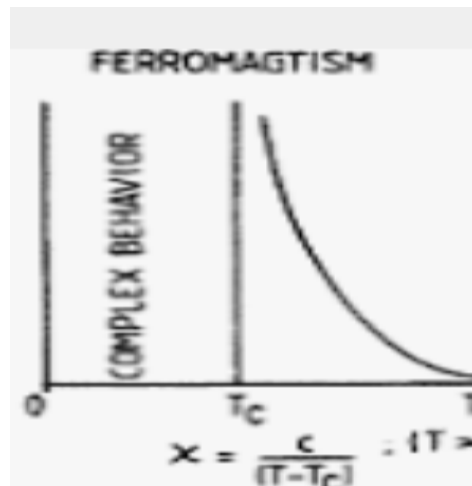
Spin alignment of ferromagnetic materials

Properties of ferro magnetic materials

The ferromagnetic material can be easily magnetized

The magnetic susceptibility is positive and depends upon

temperature $\chi = \frac{C}{T - T_C}$



When the temperature of the material is greater than its curie temperatures then it's converted into paramagnetic material

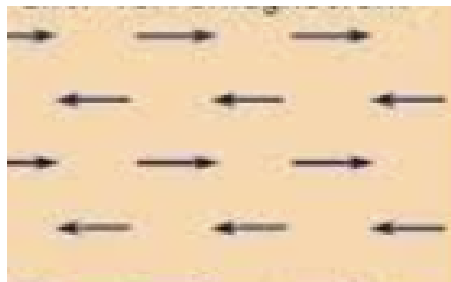
The relative permeability also very high

These materials exhibit a hysteresis curve that is B -H curve the alignment of the atoms is parallel to other in the same direction

Examples are iron Cobalt Nickel

Anti-Ferro magnetic material

This is the property of a substance whose spin movement of neighboring atoms are oriented in anti-parallel order that is the alternative atoms have spin moment parallel to each other, one substrate is spontaneously magnetized in one direction and the other in the opposite direction, if no external field is applied the net magnetization of antiferromagnetic material will be zero because of anti-parallel spin magnetic moments, and applied external magnetic field they possess magnetization



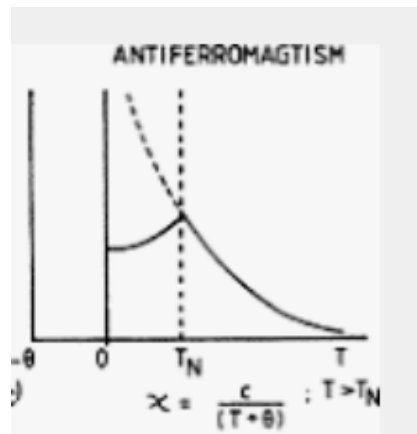
Spin alignment of anti-ferromagnetic materials

Properties of anti-ferro magnetic materials

The anti-ferromagnetic material can be easily magnetized

The magnetic susceptibility is positive and depends upon

temperature $\chi = \frac{C}{T + T_N}$



The spin alignment is the anti-parallel manner

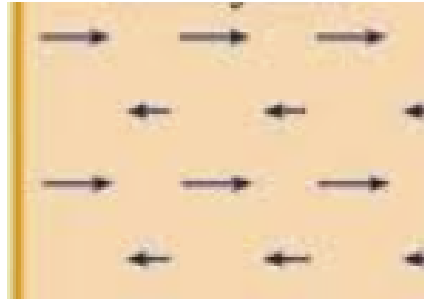
the relative permeability is greater than 1

the temperature at which the anti-ferromagnetic materials is converted into paramagnetic material is known as Neels temperatures

examples Cr_2O_3 , MnF_2 Mno, Mns

Ferrimagnetic material(ferrites)

This kind of materials is similar to ferromagnetic materials, the magnetic moments of these materials are unequal spin moment the neighboring atoms in ferric differ in the magnitude of magnetic moments and their directions also, thus that net magnetization of ferric has non zero values but small they possess a small value of magnetization



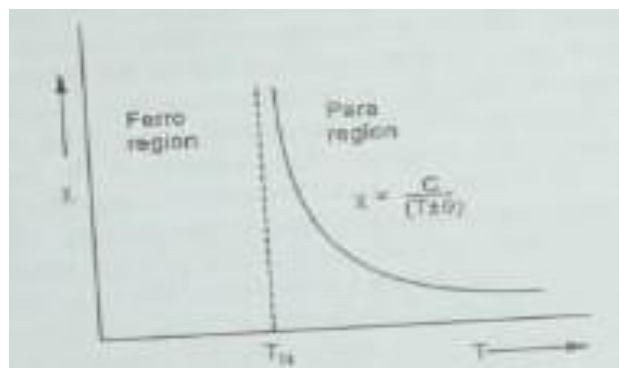
Spin alignment of anti-ferromagnetic materials

Properties of Ferrimagnetic material

The Ferrimagnetic material can be easily magnetized

The magnetic susceptibility is positive and depends upon

temperature $\chi = \frac{C}{T \pm T_N}$



The relative permeability is small

The spin alignments systematic but in anti-parallel manner

Examples Nickel ferrite, and Cobalt ferrite

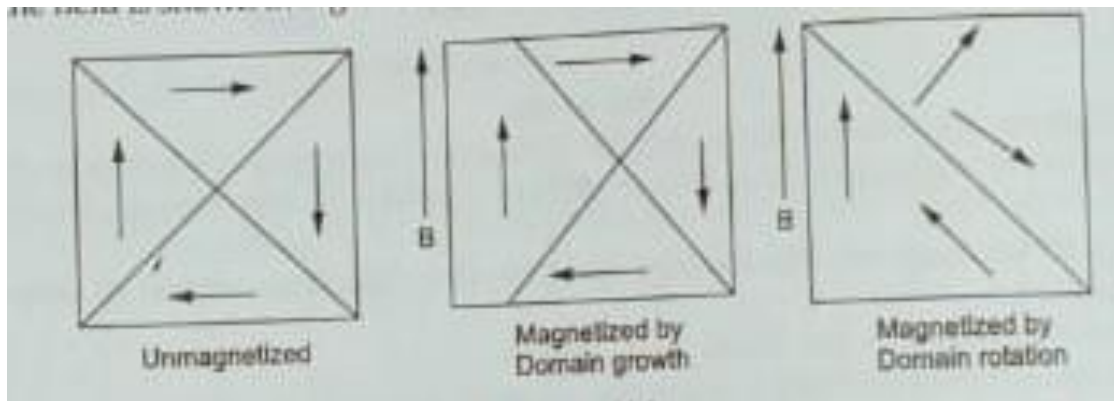
Weiss domains theory

The phenomenon of magnetic domain was first explained by Weiss, according to him ferromagnetic materials are divided into a large number of small regions is called domains.

The magnetic dipoles in a domain are aligned parallel to each other in the same direction the magnetic dipole is different in different directions. when the magnetic substance is placed in an external

magnetic field then there may be two independent processes that contribute to the total magnetization of substance

The first one is when a small magnetic field is applied to the domains which are oriented to the field direction and it is the expands of other domain which are not in field directions this gives only small magnetization

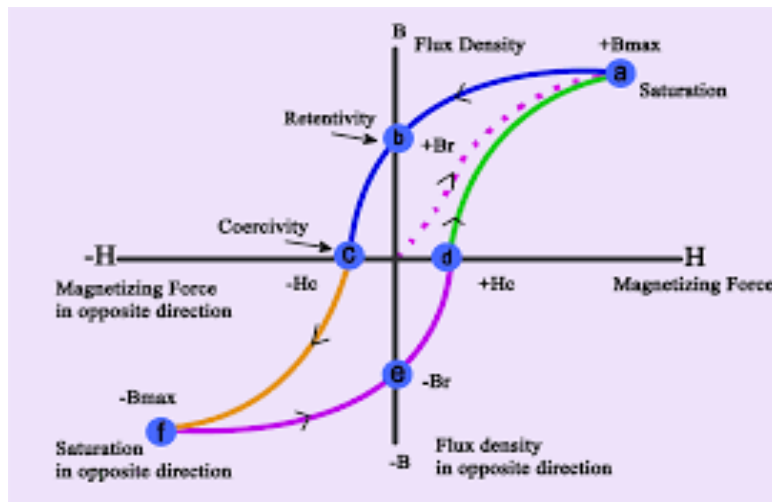


The strong field is applied to a domain and oriented in the field direction by rotation of their dipoles the resulting magnetization is large, if their strength of the magnetic field is increased the specimen gets fully magnetizations the difference between these two effects is simultaneously shown in above figure

When a small magnetic field is applied across the specimen the domain is an expand and give small magnetization but the displacement of boundaries is reversible within the regions as the external field is increased as a large number of domains grow favorable in the direction of the external field and also increase magnetization but its displacement is invisible the specimen at this moment is completely magnetized

HYSTERESIS CURVE (B-H CURVE)

A great deal of information can be learned about the magnetic properties of a material by studying its hysteresis loop in the below figure.



This hysteresis loop shows the relationship between the induced magnetic flux density that is magnetic induction (B) and the magnetizing force that is magnetic field intensity (H). It is often referred to as the B-H curve.

The loop is generated by measuring the magnetic flux (B) of ferro magnetic materials while the magnetic field intensity (H) is changed.

Let us consider a magnetic substance is placed in an external magnetic field. The substance is slowly magnetized by magnetic field intensity. The intensity of magnetization is found to increase along with O to A and decreasing H slowly B also decreases but does not follow an original path. When H is zero, B is not zero. The location of B is called retentivity.

When the field is applied in reverse direction, it decreases along B to C till it's become zero at C . The value of H which has to be applied to the magnetic material in the reverse direction the magnetization is zero. The location of C is called Coercivity.

Based on the hysteresis loop area the magnetic materials are classified under two types Soft magnetic materials and Hard magnetic materials and from the hysteresis loop a number of primary magnetic properties of a material can be determined

Retentivity -The property of the magnetic material to retain magnetism even in the absence of the magnetizing field is known as retentivity

Residual Magnetism or Residual flux

The magnetic flux density that remains in a material when the magnetizing force is zero. Not that residual magnetism and retentivity are the same when the material has been magnetized to the saturation point. However, the level of residual magnetism may be lower than the retentivity value when the magnetizing force did not reach the saturation point

Coercivity- The magnetizing field (H) needed to demagnetize the magnetic material completely is known as its coercivity

Difference between soft and hard magnetic materials

Soft magnetic materials	Hard magnetic materials
Have small hysteresis loop	Have large hysteresis loop
Have large values of Permeability and Susceptibility	Have small values of Permeability and Susceptibility
The domain wall moment is easier	The domain wall moment is difficult
Coercivity and Retentivity is small	Coercivity and Retentivity is large
Can be easily magnetized and the demagnetized	Cannot be easily magnetized and the demagnetized
Magneto-static energy is small	Magneto-static energy is large
Eddy current loss is more due to high resistivity	Eddy current loss is less due to low resistivity
Used to make electromagnets	Used to make permanent magnets
Exp: Iron -silicon-alloys ferrous-nickel- alloys	Exp: Iron Nickel -aluminum alloys copper Nickel -iron alloys

Applications of Eddy Current

1. Implemented in trains those have eddy current brakes
2. Used to offer damping torque in PMMC devices
3. Utilized in electrical devices such as induction type energy meters
4. These are employed to know damages in the metal sections.
5. Simple high frequency eddy current testers have been used for to detect toe cracks.
6. Automated eddy current test systems have been developed for inspecting tube, bar and wire at speeds up to 3 m/s

Problems-1

A magnetic field of 1800 amp/m produces a magnetic flux of $3 \times 10^{-5} \text{ Wb/m}^2$ in an iron bar of cross-sectional area 0.2 cm^2 , calculate permeability

$$\text{Formula } \mu = \frac{B}{H}$$

Problems-2

Find the relative permeability of a ferromagnetic material field strength to 220 amp/m produces a magnetization 300 m/ in it

$$\text{Formula } \mu_r = (1 + x) \text{ ----- } x = \frac{M}{H}$$

Problems-3

A magnetic material has a magnetization of 3000 amp /m flux density of 0.005w/m² calculate the magnetic force and the relative permeability of material

$$\text{Formula } H = \frac{B}{\mu_0} - 1$$

$$\text{Formula } \mu_r = (1 + \chi) \text{ ----- } \chi = \frac{M}{H}$$

Problems-4

A paramagnetic material has a magnetic field intensity of 10⁴ amperes. If the susceptibility of materials at room temperature is 3.7×10^{-3} · calculate the magnetization and flux density of metals

$$\text{Formula } \chi = \frac{M}{H} \quad M = \chi H$$

$$\text{Formula } B = \mu_0(M + H)$$

Unit-4 Quantum Mechanics

part-A

Quantum Mechanics: Dual nature of matter –Heisenberg's Uncertainty Principle –Physical Significance and properties of wave function –Schrodinger's time independent and dependent wave equations–Particle in a one-dimensional infinite potential well

Introduction

Concept of particle and wave

Particle

particle is a physical object

particle has some mass

it occupies a small area in a space

it is a small area in a space

it is easily to locate the position of a particle

it is specified by displacement velocity and momentum

Wave

wave is nothing but disturbance produced in the media

it is move from one place to another place

it occupies a very large area in a space

it is very difficult to locate the position of the wave in a space

it is it spread by wavelength frequency and space

Dual nature of light

Light has a dual nature, sometimes it behaves like a particle (called a photon), which explains how light travels in straight lines, sometimes it behaves like a wave, which explains how light bends around an object

Scientists accept the evidence that supports this dual nature of light.

Light is thought to consist of tiny bits of energy that behave like particles called photons, and Electromagnetic energy travels in waves, the waves are different from each other in their lengths are called wavelength.

we know that the light wave showing particle nature and wave nature is known as dual nature

for example, the phenomenon of interference diffraction and polarization are successfully explained by the wave nature of light, Similarly, the phenomenon of the Photoelectric effect Compton effect and blackbody radiation are successfully explained by the particle nature of light.

In 1924 the de-Broglie concludes that the light beam exhibits dual nature of light, according to him a particle is accelerated by an external field then it will spread like a wave with a certain wavelength that wave is associated with moving particle.

Their wavelength is $\lambda = \frac{h}{mv}$

Here h is Planck's Constant

m is mass of the particle

v is velocity of the particle

the de-Broglie waves are known as matter waves

Derivation of de Broglie wavelength

from the Particle nature of light according to Planck's the energy of the photon is

$$E = h\nu \text{-----(1)}$$

h is Planck's Constant

ν is Frequency of radiation

Similarly, according to Einstein, the energy of the photon of mass can be expressed as

$$E = mc^2 \text{-----(2)}$$

m is mass of the particle

c is velocity of the light

From eqe 1 and 2 and we get

$$h\nu = mc^2$$

$$h = \frac{c}{\lambda} = mc^2 \text{ since } \nu = \frac{c}{\lambda}$$

$$\lambda = \frac{h}{mc}$$

$$\lambda = \frac{h}{mv} \text{ this is called de-Broglie wavelength}$$

Expression for de- Broglie wavelength in terms of kinetic energy

We know that the kinetic energy $E = \frac{1}{2}mv^2$

Multiply with **m** on both side of the above equation

$$mE = \frac{1}{2}mv^2$$

$$2mE = m^v v^v$$

$$mv = \sqrt{2mE}$$

We know that $\lambda = \frac{h}{mv}$

$$\lambda = \frac{h}{\sqrt{2mE}}$$

The wavelength associated with moving particles and accelerated by a potential v and charged particle carrying charge q is accelerated through the potential differences v

Energy (E) = charge (q) $\times$ the potential difference (v)

$$\lambda = \frac{h}{\sqrt{2mqv}}$$

$$\lambda = \frac{6.626 \times 10^{-34}}{\sqrt{2V \times 1.6 \times 10^{-19} \times 9.11 \times 10^{-31}}}$$

$$\Rightarrow \lambda = \frac{12.27 \times 10^{-10}}{\sqrt{V}} \text{ meter}$$

$$(\text{or}) \lambda = \frac{12.27}{\sqrt{V}} \text{ \AA}$$

Properties of matters waves

Smaller the mass of the particle greater is the wavelength associated with it

The velocity of the matter waves depends up on the velocity of the material particle.

The velocity of the matter waves can be greater than the speed of light.

Matter waves are not electromagnetic in nature

Matter-wave represents the probability of finding a particle in space.

Matter waves are independent of the charge on the material particle.

Electron microscope works on the basis of de-Broglie waves.

Matter waves can propagate in a vacuum; hence they are not mechanical wave.

Heisenberg Uncertainty Principle

In 1927 Heisenberg proposed a very interesting principle known as the Uncertainty Principle which states that it is impossible to determine both the exact position and the exact momentum of a moving particle at the same time, in general,

According to classical theory, it is possible for a particle to occupy a fixed position and have a definite Momentum, therefore it is possible to predict neatly its position and momentum.

According to the Uncertainty Principle, it is not possible to determine accurately the simultaneous values of position and momentum of a particle at any given time

hence Heisenberg Uncertainty Principle states that the measurement of physical quantities which are impossible to measure accurately according to classical theory may find as the probability of occurrence

Salient features of Heisenberg Uncertainty Principle

if Δx denote the error in the measurement of the position of the particle

and represent the error in the measurement of momentum Δp

$$\text{therefore } (\Delta x)(\Delta p) = h$$

Where it is a Planck's Constant (6.6×10^{-34} joule-second)

The above relation represents the extent of the uncertainty involved in the measurement of both the position and momentum of the particle.

when one defines Δx and Δp as rms derivation and assumes the optimum measurements produce.

the above equation can be written as

$$(\Delta x)(\Delta p) \geq \frac{\hbar}{2} \quad \hbar = \frac{h}{2\pi}$$

and other form of uncertainty concerns energy and time is

$$(\Delta x)(\Delta t) \geq \frac{\hbar}{2}$$

This gives Uncertainty in measurement of energy and time of the process

Schrodinger time-independent wave equation

In 1926 Schrodinger was studied the relation between the momentum of particle and momentum of the wave and he is it developed a new theory known as wave mechanics as a part of wave mechanics is the developed two equations that is time Independent and time-dependent Wave equations.

Let consider a system of waves associated with material particles of mass **m** moving with velocity **v** is associated with wave function **ψ**

let x y z be the coordinators of the particle and be the wave disturbance for the de Broglie wave at any time **t** and **ψ** is called the wave function

the differential equation of wave motion is given by

$$\frac{\partial^2 \psi}{\partial t^2} = V^2 \left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} \right)$$

$$\frac{\partial^2 \psi}{\partial t^2} = V^2 \nabla^2 \psi \text{-----(1)} \quad \text{since } \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

∇^2 = Laplace operator

$$\psi = \psi_0 \sin \omega t$$

$$\omega = 2\pi \nu$$

ω = wave velocity

$$\psi = \psi_0 \sin 2\pi \nu t$$

The above equation differentiating with respect to **t**

$$\frac{\partial \psi}{\partial t} = \psi_0 \cos 2\pi \nu t (2\pi \nu)$$

Again, differentiating with respect to **t**

$$\frac{\partial^2 \psi}{\partial t^2} = -\psi_0 \sin 2\pi \nu t (4\pi^2 \nu^2)$$

$$= -4\pi^2 \nu^2 \psi \quad \text{since } \psi = \psi_0 \sin 2\pi \nu t$$

We know that $\vartheta = \frac{V}{\lambda}$

$$\frac{\partial^v \psi}{\partial t^v} = -4\pi^2 \frac{V^v}{\lambda^v} \psi \text{ --- (2)}$$

From eqe 1 and 2 we get

$$V^v \nabla^v \psi = -4\pi^2 \frac{V^v}{\lambda^v} \psi$$

$$\nabla^v \psi + \frac{4\pi^2}{\lambda^v} \psi = 0 \text{ -----(3)}$$

We know that $\lambda = \frac{h}{mv}$

$$\lambda^v = \frac{h^v}{m^v v^v}$$

$$\nabla^v \psi + \frac{4\pi^2 m^v v^v}{h^v} \psi = 0 \text{ -----(4)}$$

If E is total energy, then

$$E = P E + K E$$

$$E = V + \frac{1}{2} m v^v$$

$$E - V = \frac{1}{2} m v^v$$

Multiply with **m** on both side of the above equation

$$m (E - V) = \frac{1}{2} m^v v^v$$

$$2m (E - V) = m^v v^v \text{ -----(5)}$$

Now eqe 5 in 4 we get

$$\nabla^2 \psi + \frac{4\pi^2 2m (E - V)}{h^2} \psi = 0$$

$$\nabla^2 \psi + \frac{8\pi^2 m (E - V)}{h^2} \psi = 0$$

$$\nabla^2 \psi + \frac{8\pi^2 m}{h^2} (E - V) \psi = 0 \text{ -----(6)}$$

This is called Schrodinger time-independent wave equation

$$\text{We know that } \hbar = \frac{h}{2\pi} \text{ and } \hbar^2 = \frac{h^2}{4\pi^2}.$$

$$\nabla^2 \psi + \frac{2m}{\hbar^2} (E - V) \psi = 0 \text{ -----(7)}$$

For free particle $V=0$ hence the Schrodinger time-independent wave equation

$$\nabla^2 \psi + \frac{2m}{\hbar^2} E \psi = 0 \text{ -----(8)}$$

Schrodinger time-dependent wave equation

The motion of wave function can be understood by the wave function ψ which can be expressed as

$$\psi = \psi_0 e^{-i\omega t} \text{-----}(1)$$

The above equation differentiating with respect to t

$$\frac{\partial \psi}{\partial t} = \psi_0 e^{-i\omega t} (-i\omega)$$

We know that

$$\frac{\partial \psi}{\partial t} = \psi_0 e^{-i\omega t}$$

So above eqe can be written as

$$\frac{\partial \psi}{\partial t} = -i\omega \psi \quad \text{since } \omega = 2\pi\nu$$

$$\frac{\partial \psi}{\partial t} = -i2\pi\nu \psi \text{-----}(2)$$

We know that $E = h\nu$

$$\nu = \frac{E}{h} \text{-----}(3)$$

Eqe 3 in 2 we get

$$\frac{\partial \psi}{\partial t} = -i2\pi \frac{E}{h} \psi \quad \text{since } \hbar = \frac{h}{2\pi}$$

$$\frac{\partial \psi}{\partial t} = \frac{-iE}{\hbar} \psi \quad \text{since } \frac{-1}{i} = i$$

$$E\psi = -\frac{\hbar}{i} \left(\frac{\partial \psi}{\partial t} \right) \quad i^2 = -1$$

$$E\psi = i\hbar \left(\frac{\partial \psi}{\partial t} \right) \text{-----}(4)$$

From Schrodinger time-independent wave equation

$$\nabla^2 \psi + \frac{8\pi^2 m}{h^2} (E - V) \psi = 0$$

$$\nabla^2 \psi = \frac{-8\pi^2 m}{h^2} (E - V) \psi$$

$$\nabla^2 \psi = \frac{-8\pi^2 m}{h^2} E\psi - V\psi$$

$$E\psi - V\psi = \frac{-\nabla^2 \psi h^2}{8\pi^2 m}$$

$$E\psi = \left(V\psi - \frac{\nabla^2 \psi h^2}{8\pi^2 m} \right)$$

$$E\psi = \left(V - \frac{\nabla^2 h^2}{8\pi^2 m} \right) \psi \quad \text{since } E\psi = i\hbar \left(\frac{\partial \psi}{\partial t} \right)$$

$$i\hbar \left(\frac{\partial \psi}{\partial t} \right) = \left(V - \frac{\nabla^2 h^2}{8\pi^2 m} \right) \psi \quad \text{-(5) since } V - \frac{\nabla^2 h^2}{8\pi^2 m} = H$$

$$E\psi = H\psi$$

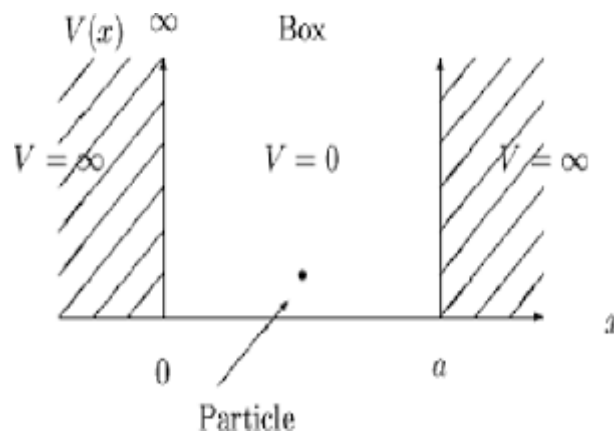
This called Schrodinger time-dependent wave equation

E is Energy operator

H is Hamiltonian operator

Particle in 1-Dimensional Potential Box

Let us consider a particle of mass m in a one-dimensional potential box moving along the x -axis between the two hard walls the box is extended from $x = 0$ to $x = a$ that is the width of the box as shown in below figure



The potential energy of particle between the two hard walls is constant because no force is acting on the particle and also the particle will not lose any energy with collisions and hence the total energy remains constant

so inside the box we can take the potential energy $V = 0$

the potential function is defined in the following way

$$V = 0 \text{ when } 0 < x < a$$

$$V = \infty \text{ when } x \leq 0 \text{ and } x \geq a$$

Now applying the Schrodinger wave equation for inside the box of the particle is given by

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2} (E - V) \psi = 0 \text{---(1)}$$

Inside the box the potential energy $V = 0$, so above eqn can be written as

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2} E \psi = 0 \text{-----(2)}$$

where $\frac{8\pi^2 m E}{h^2} = K^2$

$$\frac{d^2\psi}{dx^2} + K^2\psi = 0 \text{-----(3)}$$

General solution of ψ can be written as

$$A \sin K(x) + B \cos K(x) \text{-----(4)}$$

Where A and B are arbitrary constants to find these values, we are applying boundary conditions

Since the particle cannot penetrate to the high potential barriers

When $x=0$, $\psi = 0$ since $V(x)$ is infinite at $x=0$

When $x=a$, $\psi = 0$ since $V(x)$ is infinite at $x=a$

Substituting the above two boundary conditions in eq 4

$$0 = A \sin K(0) + B \cos K(0)$$

$$0 = 0 + B$$

$$B = 0$$

$$0 = A \sin K(a) + 0$$

$$A \sin K(a) = 0$$

$$A \neq 0$$

$$\sin K(a) = 0$$

$$K(a) = \pm n\pi$$

$$K = \frac{n\pi}{a}$$

$$k^2 = \frac{n^2 \pi^2}{a^2} \quad \text{since} \quad \frac{8\pi^2 m}{h^2} = K^2$$

$$\frac{8\pi^2 m E}{h^2} = \frac{n^2 \pi^2}{a^2}$$

$$E = \frac{n^2 h^2}{8ma^2} \text{-----(5)}$$

The above equation indicates particle in a one-dimensional potential box can take discrete values of energies for $n=1,2,3, \dots$ so, the energy is the quantized

The value of **A** calculated by using the normalizing process the probability of finding the particle inside the box along x-direction extraction

Find the A value

$$\psi = A \sin \left(\frac{n\pi}{a} \right) x$$

$$\int_{-\infty}^{+\infty} A^2 \sin^2 \left(\frac{n\pi}{a} \right) x dx = 1 \quad \text{since} \quad \int_{-\alpha}^{+\alpha} |\psi|^2 dx = 1$$

$$A^2 \int_{-\alpha}^{+\alpha} \left[1 - \frac{\cos 2 \left(\frac{n\pi}{a} \right) x}{2} \right] dx = 1$$

$$\frac{A^2}{2} \int_{-\infty}^{+\infty} \left[x - \frac{\sin 2 \left(\frac{n\pi}{a} \right) x}{2 \left(\frac{n\pi}{a} \right)} \right]_0^a = 1$$

$$\frac{A^2}{2} [a - 0] - [0 - 0] = 1$$

$$\frac{A^2}{2} [a] = 1$$

$$A^2 = \frac{2}{a}$$

$$A = \sqrt{\frac{2}{a}}$$

$$\psi = \sqrt{\frac{2}{a}} \sin \left(\frac{n\pi}{a} \right) x$$

Calculate the Eigen values

The discrete values of energies are called Eigenvalues and their functions are also called eigenfunctions and n is called the quantum number

We know that $E = \frac{n^2 h^2}{8ma^2}$

Case-1

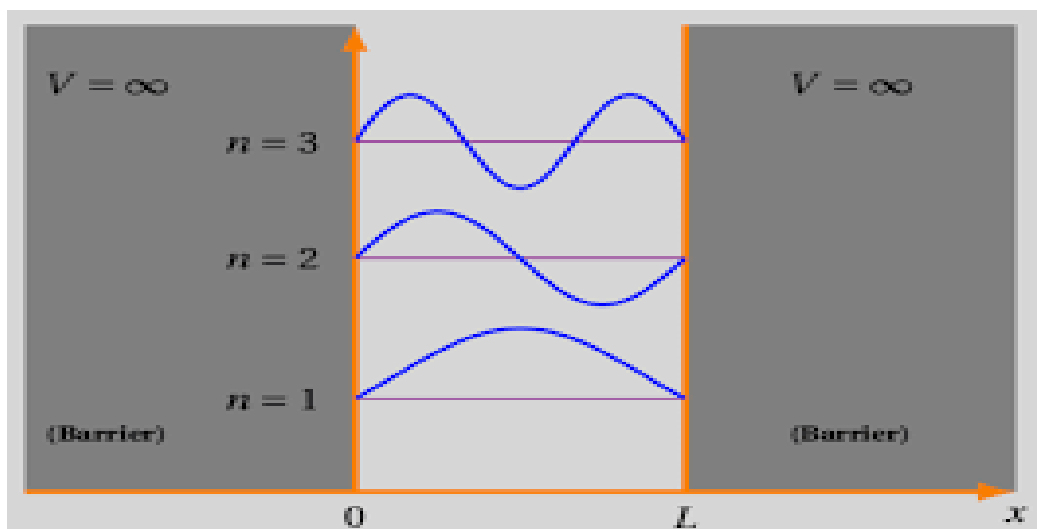
If $n=1$ then $E = \frac{h^2}{8ma^2}$ $\psi = \sqrt{\frac{2}{a}} \sin \left(\frac{\pi x}{a} \right)$

Case-2

$n=2$ then $E = \frac{4h^2}{8ma^2}$ $\psi = \sqrt{\frac{2}{a}} \sin \left(\frac{2\pi x}{a} \right)$

Case-3

$n=3$ then $E = \frac{9h^2}{8ma^2}$ $\psi = \sqrt{\frac{2}{a}} \sin \left(\frac{3\pi x}{a} \right)$



Physical significance of(ψ) wave function.

The wave function (ψ) has no direct physical meaning

ψ^* is a complex conjugate of ψ

ψ^{**} is the probability density function

The first derivation of must be continuous

ψ connect particle nature and associated with wave nature

ψ must be finite at value for all values of x y z

The normalized condition for the wave function for the motion of the particle can be written as

$$\int_{-\alpha}^{+\alpha} |\psi|^2 dx = 1$$

PROBLEM-1

Electron is confined to a box of uncertainty in position is 10^{-8} metre then calculate the uncertainty in its velocity

$$\text{FORMULA } \Delta x \cdot \Delta p = \frac{\hbar}{2} \quad \text{or } \Delta x \cdot m \Delta v = \frac{\hbar}{2}$$

PROBLEM-2

Calculate The wavelength associated with an electron of energy 2000

$$\text{FORMULA } \lambda = \frac{h}{\sqrt{2mE}}$$

PROBLEM-3

An electron is confined to a one-dimensional potential box of length $2A^0$. calculate the energies corresponding to the second and fourth Quantum states

$$\text{FORMULA } E = \frac{n^2 h^2}{8ma^2}$$

PROBLEM-4

Calculate The velocity of an electron of wavelength $1.66 \times 10^{-19}m$

$$\text{FORMULA } \lambda = \frac{h}{mv}$$

Unit-5 classical free electron theory

part-a

Free Electron Theory: Classical free electron theory (Qualitative with discussion of merits and demerits) – Quantum free electron theory–Equation for electrical conductivity based on quantum free electron theory-Fermi Dirac Distribution-Density of states (3D) -Fermi energy.

Classical free electron theory

The classical free electron theory was developed by Drude and Lorentz and hence is also known Drude - Lorentz theory According to this theory a metal consists of electrons that are free to move about in the Crystal-like molecules of gas in a container. mutual repulsion between the electrons is ignored and hence potential energy is taken as zero. therefore, the total energy of an electron is equal to the kinetic energy of gases, this means electrons can be assigned a mean free path mean collision time and average mean velocity, relaxation time

Salient features

The free electrons in all metals behave as the molecules of gas in a container

As the electron moves randomly, the resultant velocity of them in a particular direction will be zero

If the potential difference is created across the metal, then the free electrons slowly drift towards the positive potential or in the opposite direction to the applied electric field.

In the absence of an electric field, the free electrons move in random directions making elastic collisions with no loss of energy.

When an external field is applied, the free electrons are slowly drifting towards the positive potential then electrons are accelerated and get current

Drift velocity (V_d)

Drift velocity is defined as the velocity achieved by the charged particle in the presence of an external field is known as drift velocity

$$V_d \propto E$$

$$V_d = uE \quad u \text{ is mobility}$$

$$u = \frac{V_d}{E} \text{-----(1)}$$

We know that due to applied field electrons are accelerated

$$a = \frac{eE}{m}$$

$$\frac{dV_d}{dt} = \frac{eE}{m}$$

$$dV_d = \frac{eE}{m} dt$$

integrating on both side

$$V_d = \frac{eE}{m} t \quad t = \tau$$

$$V_d = \frac{e\tau}{m} E \text{-----(2)}$$

$$u = \frac{e\tau}{m}$$

$$\text{We know that } \sigma = \left(\frac{ne^2\tau}{m} \right)$$

$$\sigma = \left(\frac{nee\tau}{m} \right)$$

$$\sigma = ne u$$

Mean collision time (τ_c)

The mean time spent between two successive collisions is known as mean collision time

$$\tau_c = \frac{\lambda}{\bar{c}}$$

Mean free path (λ)

The mean distance travel between two successive collisions is known as mean free path

$$\lambda = \tau_c \bar{c}$$

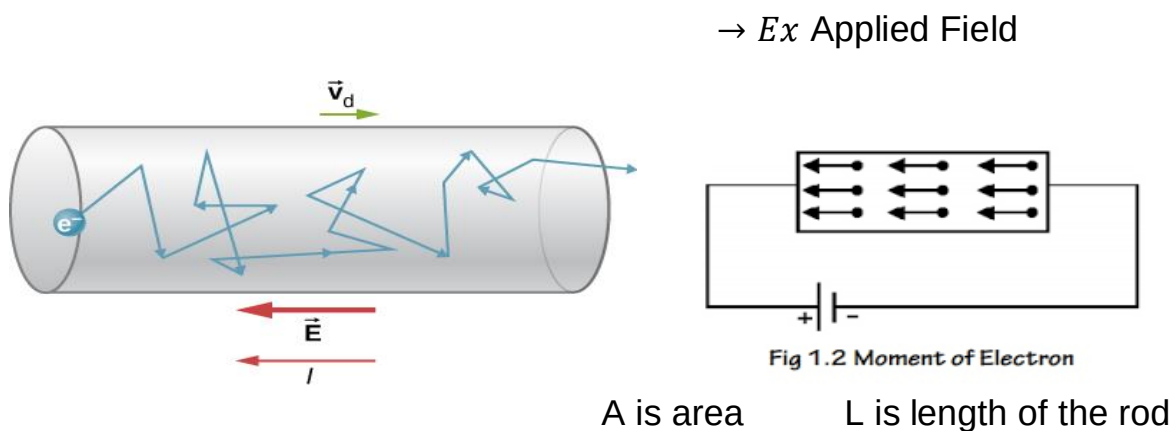
Relaxation time ' τ '

The time is taken by the electron to decrease its velocity to $1/e$ of its initial value is called relaxation time

$$\langle vx \rangle = \langle vx_0 \rangle e^{-t/\tau}$$

Expression for electrical conductivity of metals through classical free electron theory

To obtain an expression for electrical conductivity let us consider a metallic rod of length L and area of the cross-section A as shown in the Below figure



In the absence of an external electric field, the electrons are in a random direction similar to the gas molecules in a container. So that there is no

resultant motion of an electron in any direction. the root mean square of the velocity of an electron can be obtained by applying the kinetic theory of electron gas

according to kinetic theory of gas can be written as

$$\bar{C} = \sqrt{\frac{3k_B T}{m}}$$

In the absence of an external field the free electrons in a random motion so net velocity is zero and current also zero, but the external field is applied to the metals the electrons are accelerated

it can be written as

$$\begin{aligned} a &= \frac{-eE}{m} \\ \frac{dv}{dt} &= \frac{-eE}{m} \\ \frac{d\langle vx \rangle}{dt} &= \frac{-eE}{m} \text{-----(1)} \end{aligned}$$

If **n** is the number of conduction electrons for one unit volume and **e** charge of the electron

then-current density in terms of average velocity is given by

$$J = n \langle vx \rangle (-e) \text{-----(2)}$$

Ege 2 diff w r t t

$$\begin{aligned} \frac{dJx}{dt} &= n \frac{d \langle vx \rangle}{dt} (-e) \\ \frac{dJx}{dt} &= n \frac{-eE}{m} (-e) \\ \frac{dJx}{dt} &= \frac{ne^2 E}{m} \\ \frac{dJx}{dt} &\propto E \text{-----(3)} \end{aligned}$$

From above result it is failed to ohm's law

But classical free electron theory obeys the ohm's law that is

$$J = \sigma E$$

To satisfy the ohms law de-acceleration term is introduce

It can be written as

$$\left[\frac{d\langle vx \rangle}{dt} \right]_{\text{Applied field}} + \left[\frac{d\langle vx \rangle}{dt} \right]_{\text{lattice collision}} = 0 \text{-----(4)}$$

$$\langle vx \rangle = \langle vx_0 \rangle e^{-t/\tau} \text{-----(5)}$$

Ege 4 diff w r t t

$$\frac{d\langle vx \rangle}{dt} = \langle vx_0 \rangle e^{-t/\tau} - 1/\tau$$

$$\frac{d\langle vx \rangle}{dt} = \frac{-\langle vx \rangle}{\tau} \text{-----(6)}$$

From ege 1 and 6 in 4 we get

$$\frac{-eE}{m} - \frac{\langle vx \rangle}{\tau} = 0$$

$$\frac{-eE}{m} = \frac{\langle vx \rangle}{\tau}$$

$$\langle vx \rangle = \frac{-eE\tau}{m} \text{-----(7)}$$

We know that $J = n \langle vx \rangle (-e)$

$$J = n \frac{-eE\tau}{m} (-e)$$

$$J = \frac{ne^2 E \tau}{m}$$

$$J = \left(\frac{ne^2\tau}{m} \right) E$$

$$J \propto E$$

We know that ohm's law $J = \sigma E$

$$\sigma E = \left(\frac{ne^2\tau}{m} \right) E$$

$$\sigma = \left(\frac{ne^2\tau}{m} \right)$$

$$\tau = \frac{\lambda}{\bar{c}}$$

$$\sigma = \left(\frac{ne^2\lambda}{m\bar{c}} \right) \quad \text{since } \bar{c} = \sqrt{\frac{3k_B T}{m}}$$

$$\sigma = \left(\frac{ne^2\lambda}{m\sqrt{\frac{3k_B T}{m}}} \right)$$

$$\sigma = \left(\frac{ne^2\lambda}{\sqrt{3mk_B T}} \right)$$

Merits and Demerits

Merit ::

- 1) It explains the electrical conductivity and thermal conductivity of metals.
- 2) It explains the Wiedemann - Franz law.
- 3) It verifies Ohm's law.
- 4) It is used to explain the optical properties of metals.

Demerits

- 1) It fails to explain the electric specific heat and the specific heat capacity of metals.
- 2) It fails to explain superconducting properties of metals.
- 3) It fails to explain new phenomena like photo-electric effect, Compton effect, Black body radiation, etc.
- 4) It fails to explain electrical conductivity of semiconductors or insulators
- 5) It fails to give a correct mathematical expression for thermal conductivity.
- 6) Ferromagnetism could not be explained by this theory.

Quantum free electron theory

classical free electron theory could not explain many physical properties such as specific heat, mean free path etc.

In 1928 Sommerfeld developed a new theory applying quantum mechanical concepts and Fermi-Dirac Statistics to the free electrons in the metals this theory is known as Quantum free electron theory

classical free electron theory permits all electrons to gain energy, but the quantum free electron theory permits only a fraction of electrons to gain energy.

according to the classical theory, the free electrons in metal move randomly such that the velocity is uniform in all directions.

but according to Quantum theory the free electrons occupy different energy levels present in the metals and they did not possible same energy values they can take some discrete energy values

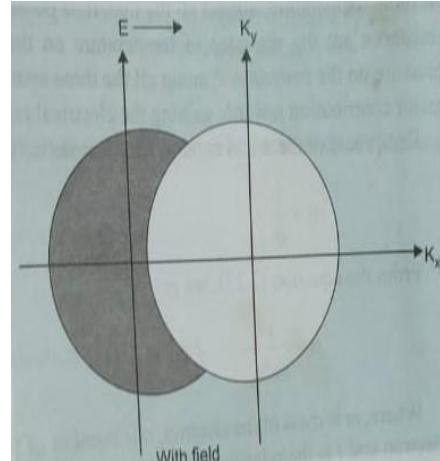
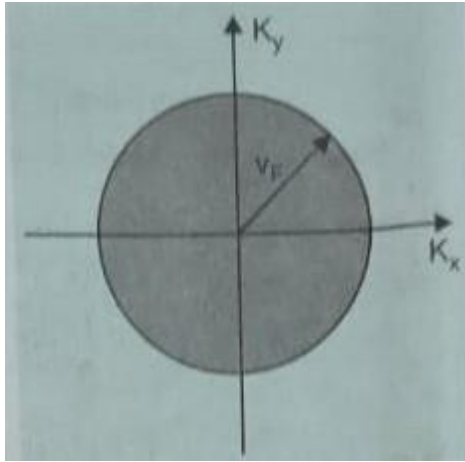
and distribute different energy states from lower energy states to the higher energy state is known as a fermi energy state

In the absence of an electric field, the free electrons move in random directions making elastic collisions with no loss of energy.

When an external field is applied, the free electrons are slowly drifting towards the positive potential then electrons are accelerated and get current

Expression for electrical conductivity of metals through Quantum free electron theory

let us consider the sphere of radius V_F is assume as shown in below fig



in which each point represented as the velocity of free electron and this sphere is called the fermi sphere, the velocities are joining different point inside the sphere from the origin represent velocities vectors

in the absence of an electric field, all electrons are move in random motion within the sphere so net velocity is zero.

if electric field E applied on the free electrons, then entire system is shifted in opposite direction to the applied field direction

according to Quantum theory the momentum

$$P = mv = \hbar k \text{-----(1)}$$

$$mv = \hbar k \quad \text{Where } \hbar = \frac{h}{2\pi} \quad k = \frac{2\pi}{\lambda}$$

$$V = \frac{\hbar k}{m} \text{-----(2)} \quad \frac{h}{\lambda} = \frac{h}{mV}$$

$$\frac{dV}{dt} = \frac{\hbar}{m} \cdot \frac{dk}{dt} \text{-----(3)}$$

We know that acceleration

$$a = \frac{eE}{m}$$

$$\frac{dV}{dt} = \frac{eE}{m} \text{-----(4)}$$

Ege 4 in 3 and we get

$$\frac{eE}{m} = \frac{\hbar}{m} \cdot \frac{dk}{dt}$$

$$dk = \frac{eE}{\hbar} dt \text{-----(5)}$$

integrating the ege 5 taking limits from 0 - t

$$\int_0^t dk = \int_0^t \frac{eE}{\hbar} dt$$

$$K(t) - K(0) = \frac{eE}{\hbar} t \quad \text{since } K(t) - K(0) = \Delta K$$

$$\Delta K = \frac{e\tau}{\hbar} E \text{-----(6)} \quad t = \tau$$

From ege 2 we can written as

$$V = \frac{\hbar k}{m}$$

$$\Delta V = \frac{\hbar}{m} \Delta K \text{-----(7)}$$

Now Ege 6 in 7 we get

$$\Delta V = \frac{\hbar}{m} \frac{e\tau}{\hbar} E$$

$$\Delta V = \frac{e\tau}{m} E \text{-----(8)}$$

If the increased velocity is ΔV due to application of electric field, then the current density

$$J = n \Delta V(e) \text{-----(9)}$$

Ege 8 in 9 and we get

$$J = n \frac{e\tau}{m} E(e)$$

$$J = \frac{n e^2 \tau}{m} E$$

$$J \propto E \quad \text{since } J = \sigma E$$

$$\sigma E = \frac{n e^2 \tau}{m} E$$

$$\sigma = \frac{n e^2 \tau}{m}$$

Merits and Demerits of Quantum free electron theory

Merits

1. This theory explains the specific heat capacity of materials.
2. This theory explains photo electric effect, Compton Effect and black body radiation. Etc
3. This theory gives the correct mathematical expression for the thermal conductivity of metal
4. this theory could explain many properties of metals

Demerits

1. This theory fails to distinguish between metal, semiconductor and Insulator.
2. It also fails to explain the positive value of Hall Co-efficient.
3. According to this theory, only two electrons are present in the Fermi level and they are responsible for conduction which is not true
4. It also fails to explain large value of mean free path

Fermi Dirac Distribution

Sommerfeld introduced a parameter called the Fermi-Dirac distribution function, this function gives the probability of occupation of an energy state by a single electron.

The probability $f(E)$ for the occupation of a particular energy level E by an electron is given by

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}$$

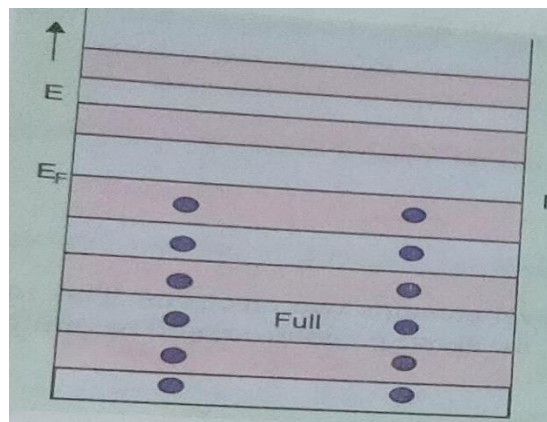
Where $f(E)$ is Fermi-Dirac Distribution

E is Energy State

E_F is Fermi Energy State

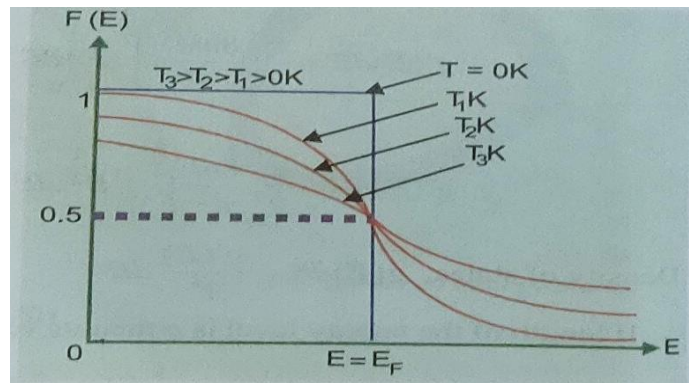
k_B is Boltzmann Constant

T is Absolute Temperature



The above diagram can represent the energy states which are separated into the filled energy states and empty energy states and at 0 K all electrons are filled below the Fermi energy state, the Fermi energy state is called the highest energy E_F .

When the temperature is increased, all electrons vacated from below the Fermi level and occupied above the Fermi level and participated in the conduction.



The above diagram represent the variation of fermi Dirac distribution with temperature

Case -1

If $E > E_F$ then $f(E) = 0$

Case -2

If $E < E_F$ then $f(E) = 1$

Case -2

If $E = E_F$ then $f(E) = \frac{1}{2}$

Fermi level

The highest energy level that an electron can occupy at the absolute temperatures is known as the Fermi level. the Fermi level is lies between the valance band and conduction band because at Absolute Temperature the electrons are all in the lowest energy state

The electrons are fermi ions and obey the parulis exclusion principle. according to which an energy level can accommodate at most two electrons. One with spin up and the other spin down.

thus, in filling the energy level two electrons occupy the lowest level two more in the next level and up to fermi level therefore the highest occupied energy level by an electron at 0K is called fermi level

Fermi energy

The energy state at which the probability of electron occupation is 1/2 at any temperature above 0K

Or

it is the level of maximum energy of the field state at 0K is known as Fermi energy.

At 0k, below the fermi level all energy state is completely filled by the electrons and above the fermi level all energy states are empty.

the probability occupancy of electron at the fermi level is half at any temperature other than 0K, therefore the maximum energy of filled state by electron at 0K is called fermi energy.

That means the energy of the state at which the probability occupation of electron is half at any temperature other than 0K is called fermi energy. It can be evaluated by

$$E_F = \left(\frac{3}{\pi}\right)^{2/3} \frac{h^2}{8m^2} n^{2/3} = 3 \cdot 62 \times 10^{-19} \text{Ev}$$

Density of states

The number of energy States for one unit volume is known as density of energy states

It is donated by $Z(E) dE$

$$Z(E) dE = \frac{\pi}{2} n(n dn) \text{---(1)}$$

We know that $E = \frac{n^2 h^2}{8ma^2}$

$$n^2 = \frac{8ma^2 E}{h^2} \text{----(2)}$$

$$n = \left[\frac{8ma^2 E}{h^2} \right]^{\frac{1}{2}}$$

eqe 2 diff w r t E

$$2n dn = \left[\frac{8ma^2}{h^2} \right] dE$$

$$n dn = \frac{1}{2} \left[\frac{8ma^2}{h^2} \right] dE$$

substitute $n dn$ values in eqe in 1

$$Z(E) dE = \frac{\pi}{2} \left[\frac{8ma^2 E}{h^2} \right]^{1/2} \frac{1}{2} \left[\frac{8ma^2}{h^2} \right] dE$$

$$Z(E) dE = \frac{\pi}{4} \left[\frac{8ma^2}{h^2} \right]^{3/2} E^{1/2}$$

We know that according to Pauli's exclusion principle each energy state occupied two electrons one with spin +1/2 and other with spin -1/2 can occupy an orbital. so, the above equation can be written as

$$Z(E) dE = \frac{\pi}{2} \left[\frac{8ma^2}{h^2} \right]^{3/2} E^{1/2}$$

Problem-1

Find the relaxation time of conduction electrons in metals of resistivity

$1.54 \times 10^{-8} \Omega$ if the metal has 5.8×10^{28} conduction electrons per m^3

Formula $\sigma = \frac{n e^2 \tau}{m}$ $\sigma = \frac{n e^2}{m p}$ $\sigma = \frac{1}{p}$

Problem-2

Calculate the mean free path of an electron in copper of density $8.5 \times 10^{28} m^{-3}$ and a resistivity $1.69 \times 10^{-8} \Omega$ given that the $m = 9.11 \times 10^{-31} kg$,

$T = 300K$ and $e = 1.69 \times 10^{-19} C$

Formula $\sigma = \left(\frac{n e^2 \tau}{m} \right)$ $\sigma = \left(\frac{n e^2 \lambda}{m \sqrt{\frac{3 k_B T}{m}}} \right)$ $\sigma = \frac{n e^2 \lambda}{\sqrt{3 m k_B T}}$

$$\sigma = \frac{1}{p} \quad p = \frac{\sqrt{3 m k_B T}}{n e^2 \lambda} \quad \lambda = \frac{\sqrt{3 m k_B T}}{n e^2 p}$$

Problem-3

For the metal having 6.5×10^{28} conduction electrons per for m^3 find the relaxation time of conduction electrons if the metal has resistivity $1.43 \times 10^{-8} \Omega$

Formula $\sigma = \left(\frac{n e^2 \tau}{m} \right)$ $\tau = \left(\frac{m}{n e^2 p} \right)$ $\sigma = \frac{1}{p}$

Problem-4

If copper density equal to $8.9 \times 10^{-8} kg$, the resistivity of copper $1.73 \times 10^{-8} \Omega$, atomic weight $M = 63.5$ calculate the mobility of electrons

Formula $u = \frac{1}{p n e}$ here **n** is number of atoms

$$n = \frac{\text{density} \times \text{avogadronumber}}{\text{atomic number}}$$

Unit-5 Semiconductors

Part-B

Semiconductors: Introduction-Intrinsic semiconductors –Density of charge carriers –Electrical conductivity –Fermi level –extrinsic semiconductors –density of charge carriers –dependence of Fermi energy on carrier concentration and temperature -Drift and diffusion currents –Einstein's equation-Hall effect –Hall coefficient – Applications of Hall effect.

Introduction

In semiconductors, The Forbidden band gap is very small as shown in the above figure. Germanium and silicon are the best examples of semiconductors. In Germanium the Forbidden band gap is of the order of 0.7 eV while in the case of silicon The Forbidden energy band gap is of the order of 1.1 eV. Actually, the semiconductor material is one whose electrical properties lie between insulators and conductors at 0K. there are no electrons in the conduction band and the valence band is completely filled. when a small amount of energy is supplied, then electrons can easily jump from valence band to conduction band this type of materials is known as insulating materials semiconductors

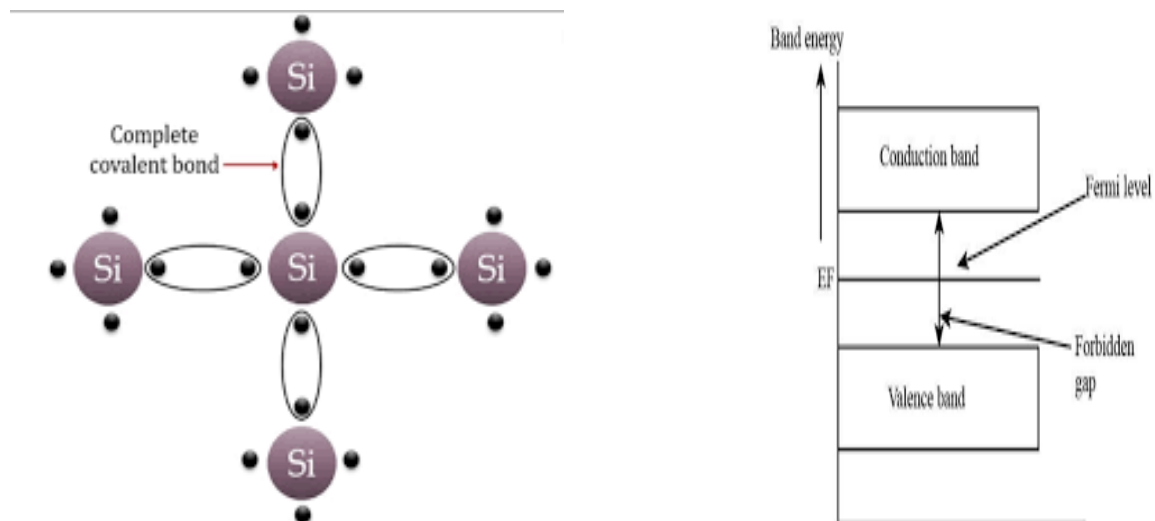
These materials are classified into two types

1. Intrinsic semi-conductors
2. Extrinsic semi-conductors
 - A) N-type semi-conductor
 - B) P-type semi-conductor

Ex- Germanium, silicon

Intrinsic semi-conductors

The semiconductor is divided into two types. The pure form of the semiconductor is known as the intrinsic semiconductor and the semiconductor in which intentionally impurities are added for making it conductive is known as the extrinsic semiconductor.



Pure Silicon and Germanium are called intrinsic semiconductors in their crystal lattice each atom with four Valence Electrons from Four covalent bonds with their adjacent atoms. when an electric field is applied to these materials the electron escape from the valency band to the conduction band. in this semiconductor, electrons and holes are always equal and the position of the Fermi energy level is exactly the midpoint of the two bands

Examples are Germanium and silicon

Extrinsic semi-conductors

an increase in the conductivity of a semiconductors when a small amount of impurity was added to it. These materials are Extrinsic Semiconductors or impurity Semiconductors. Another term for these materials is 'Doped Semiconductor.

By adding a small quantity of third and fifth group of elements as an impure into pure semiconductors these are known as extensive semiconductor the presence of adding of impurities is called doping

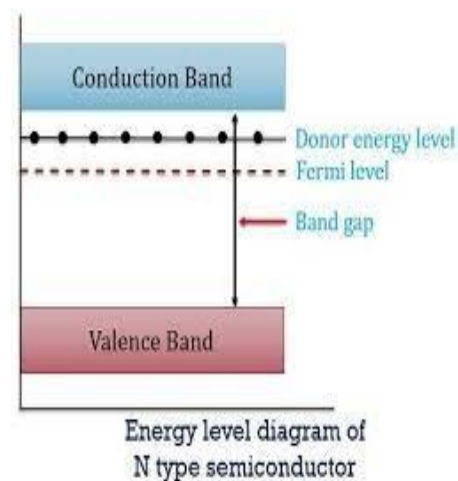
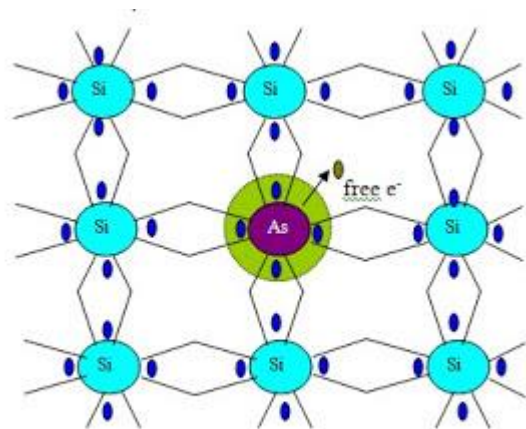
this is mainly two Types

n-type semiconductor

P-type semiconductor

n-type semiconductor

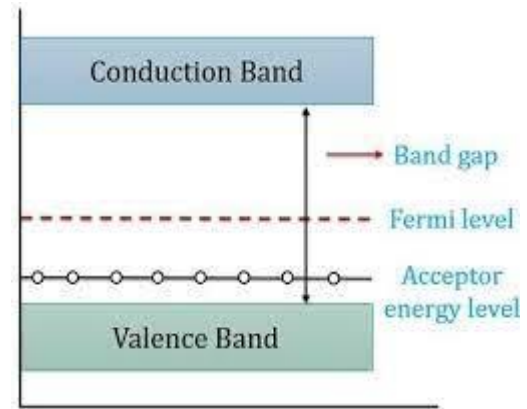
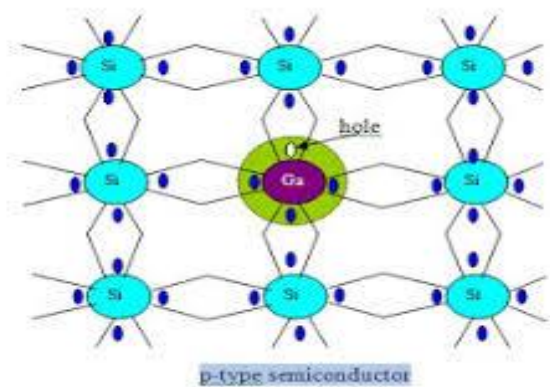
In these semiconductors the pentavalent impurities are added to pure tetravalent are called n-type semiconductors then the pentavalent impurities like arsenic is added to be silicon this impurity is called donor impurity. in this semiconductor electronics are majority charge carriers and holes are minority charge carriers and the position of Fermi energy level is very near to the conduction band



p-type semiconductor

In this semiconductor, the trivalent impurities are added to pure tetravalent are called P-type semiconductors, then the trivalent impurities like gallium is added to Silicon this type of impurities is called acceptor impurities. in this semiconductor holes are majority to charge Carriers and electrons of minority charge

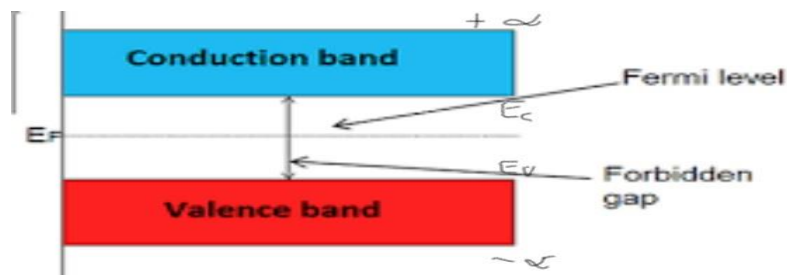
Carriers and the Fermi energy level is very near to the valency band



Density of electrons in intrinsic semiconductors

or

Carrier concentration of electrons in intrinsic semiconductors



$$n = \int_{E_C}^{\infty} Z(E) f(E) dE \text{-----(1)}$$

$$\text{here } Z(E) = \frac{\pi}{2} \left[\frac{8me^*}{h^2} \right]^{3/2} E^{1/2}$$

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}$$

$$n = \int_{E_C}^{\infty} \frac{\pi}{2} \left[\frac{8me^*}{h^2} \right]^{3/2} E^{1/2} \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)} dE$$

$$E = E - E_C \quad \text{and} \quad E - E_F \gg k_B T$$

for all possible temperature

by neglecting the first term,

$$f(E) \text{ can be written as } \exp \frac{-(E - E_F)}{k_B T}$$

$$n = \int_{E_C}^{\infty} \frac{\pi}{2} \left[\frac{8me^*}{h^2} \right]^{3/2} (E - E_C)^{1/2} \exp \frac{-(E - E_F)}{k_B T} dE$$

$$n = \int_{E_C}^{\infty} \frac{\pi}{2} \left[\frac{8me^*}{h^2} \right]^{3/2} (E - E_C)^{1/2} \exp \frac{-(E - E_C + E_C - E_F)}{k_B T} dE$$

$$n = \int_{E_C}^{\infty} \frac{\pi}{2} \left[\frac{8me^*}{h^2} \right]^{3/2} (E - E_C)^{1/2} \exp \frac{-(E - E_C)}{k_B T} \exp \frac{-(E_C - E_F)}{k_B T} dE$$

$$n = \frac{\pi}{2} \left[\frac{8me^*}{h^2} \right]^{3/2} \exp \frac{-(E_C - E_F)}{k_B T} \int_{E_C}^{\infty} (E - E_C)^{1/2} \exp \frac{-(E - E_C)}{k_B T} dE$$

$$\text{let } E - E_c = x^2 \quad dE = 2x dx$$

$$n = \frac{\pi}{2} \left[\frac{8me^*}{h^2} \right]^{3/2} \exp \frac{-(E_c - E_F)}{k_B T} \int_0^\infty (x^2)^{\frac{1}{2}} \exp\left(\frac{-x^2}{k_B T}\right) 2x dx$$

$$n = \frac{\pi}{2} \left[\frac{8me^*}{h^2} \right]^{3/2} \exp \frac{-(E_c - E_F)}{k_B T} 2 \int_0^\infty x^{\frac{1}{2}} \exp\left(\frac{-x^2}{k_B T}\right) dx$$

$$\text{Let } x^2 = t \quad x = t^{\frac{1}{2}} \quad dx = \frac{1}{2} t^{-\frac{1}{2}} dt$$

$$n = \frac{\pi}{2} \left[\frac{8me^*}{h^2} \right]^{3/2} \exp \frac{-(E_c - E_F)}{k_B T} 2 \int_0^\infty t^{\frac{1}{2}} \exp\left(\frac{-t}{k_B T}\right) \frac{1}{2} t^{-\frac{1}{2}} dt$$

$$n = \frac{\pi}{2} \left[\frac{8me^*}{h^2} \right]^{3/2} \exp \frac{-(E_c - E_F)}{k_B T} \int_0^\infty t^{\frac{1}{2}} \exp\left(\frac{-t}{k_B T}\right) dt$$

$$n = \frac{\pi}{2} \left[\frac{8me^*}{h^2} \right]^{3/2} \exp \frac{-(E_c - E_F)}{k_B T} \int_0^\infty t^{\frac{3}{2}-1} e^{-\left(\frac{1}{k_B T}\right)t} dt$$

$$\text{Since } \int_0^\infty x^{n-1} e^{-kx} dx = \frac{\Gamma n}{k^n}$$

$$\Gamma n = (n-1)\Gamma(n-1)$$

$$\Gamma n = \left(\frac{3}{2} - 1\right) \Gamma\left(\frac{3}{2} - 1\right)$$

$$\Gamma n = \frac{1}{2} \sqrt{\pi}$$

$$k^n = \left(\frac{1}{k_B T} \right)^{3/2}$$

$$\text{Therefore } \frac{\Gamma n}{k^n} = \frac{\frac{1}{2} \sqrt{\pi}}{\left(\frac{1}{k_B T} \right)^{3/2}} = \frac{1}{2} \sqrt{\pi} (k_B T)^{3/2}$$

$$n = \frac{\pi}{2} \left[\frac{8me^*}{h^2} \right]^{3/2} \exp \frac{-(E_C - E_F)}{k_B T} \frac{1}{2} \sqrt{\pi} (k_B T)^{3/2}$$

$$n = \frac{(8)^{3/2}}{4} \left[\frac{me^* \pi k_B T}{h^2} \right]^{3/2} \exp \frac{-(E_C - E_F)}{k_B T}$$

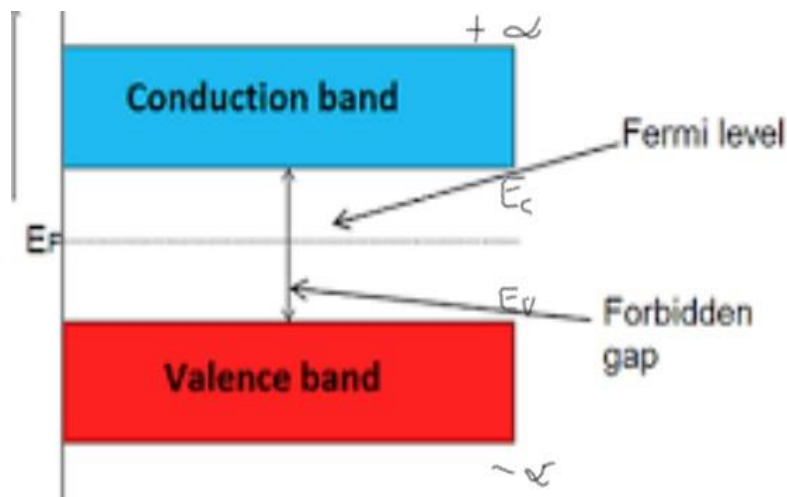
$$n = 2 \left[\frac{2me^* \pi k_B T}{h^2} \right]^{3/2} \exp \frac{-(E_C - E_F)}{k_B T}$$

$$n = N_c \exp \frac{-(E_C - E_F)}{k_B T} \quad \text{since } N_c = 2 \left[\frac{2me^* \pi k_B T}{h^2} \right]^{3/2} \text{pseudo constant}$$

Density of holes in intrinsic semiconductors

or

Carrier concentration of holes in intrinsic semiconductors



$$P = \int_{-\infty}^{E_v} Z(E) f(E) dE \text{-----}(1)$$

$$\text{here } Z(E) = \frac{\pi}{2} \left[\frac{8mh^*}{h^2} \right]^{3/2} E^{1/2}$$

$$f(E) = \frac{1}{1 + \exp\left(\frac{E_F - E}{k_B T}\right)}$$

$$P = \int_{-\infty}^{E_v} \frac{\pi}{2} \left[\frac{8mh^*}{h^2} \right]^{3/2} E^{1/2} \frac{1}{1 + \exp\left(\frac{E_F - E}{k_B T}\right)} dE$$

$$E = E_V - E \text{ and } E_F - E \gg k_B T$$

for all possible temperature

by neglecting the first term

$$f(E) \text{ can be written as } \exp \frac{-(E_F - E)}{k_B T}$$

$$P = \int_{-\infty}^{E_v} \frac{\pi}{2} \left[\frac{8mh^*}{h^2} \right]^{3/2} (E_V - E)^{\frac{1}{2}} \exp \frac{-(E_F - E)}{k_B T} dE$$

$$P = \int_{-\infty}^{E_v} \frac{\pi}{2} \left[\frac{8mh^*}{h^2} \right]^{3/2} (E_V - E)^{\frac{1}{2}} \exp \frac{-(E_F - E_V + E_V - E)}{k_B T} dE$$

$$P = \int_{-\infty}^{E_v} \frac{\pi}{2} \left[\frac{8mh^*}{h^2} \right]^{3/2} (E_V - E)^{\frac{1}{2}} \exp \frac{-(E_F - E_V)}{k_B T} \exp \frac{-(E_V - E)}{k_B T} dE$$

$$P = \frac{\pi}{2} \left[\frac{8mh^*}{h^2} \right]^{3/2} \exp \frac{-(E_F - E_V)}{k_B T} \int_{-\infty}^{E_v} (E_V - E)^{\frac{1}{2}} \exp \frac{-(E_V - E)}{k_B T} dE$$

$$\text{let } E_V - E = x^2 \quad dE = -2x dx$$

$$P = \frac{\pi}{2} \left[\frac{8mh^*}{h^2} \right]^{3/2} \exp \frac{-(E_F - E_V)}{k_B T} \int_0^\infty (x^2)^{\frac{1}{2}} \exp\left(\frac{-x^2}{k_B T}\right) - 2x dx$$

$$P = \frac{\pi}{2} \left[\frac{8mh^*}{h^2} \right]^{3/2} \exp \frac{-(E_F - E_V)}{k_B T} 2 \int_0^\infty x^{\frac{1}{2}} \exp\left(\frac{-x^2}{k_B T}\right) dx$$

$$\text{Let } x^2 = t \quad x = t^{\frac{1}{2}} \quad dx = \frac{1}{2} t^{-\frac{1}{2}} dt$$

$$P = \frac{\pi}{2} \left[\frac{8mh^*}{h^2} \right]^{3/2} \exp \frac{-(E_F - E_V)}{k_B T} 2 \int_0^\infty t^{\frac{1}{2}} \exp\left(\frac{-t}{k_B T}\right) \frac{1}{2} t^{-\frac{1}{2}} dt$$

$$P = \frac{\pi}{2} \left[\frac{8mh^*}{h^2} \right]^{3/2} \exp \frac{-(E_F - E_V)}{k_B T} \int_0^\infty t^{\frac{1}{2}} \exp\left(\frac{-t}{k_B T}\right) dt$$

$$P = \frac{\pi}{2} \left[\frac{8mh^*}{h^2} \right]^{3/2} \exp \frac{-(E_F - E_V)}{k_B T} \int_0^\infty t^{\frac{3}{2}-1} e^{-\left(\frac{1}{k_B T}\right)t} dt$$

$$\text{Since } \int_0^\infty x^{n-1} e^{-kx} dx = \frac{\Gamma n}{k^n}$$

$$\Gamma n = (n-1)\Gamma(n-1)$$

$$\Gamma n = \left(\frac{3}{2} - 1\right) \Gamma\left(\frac{3}{2} - 1\right)$$

$$\Gamma n = \frac{1}{2} \sqrt{\pi}$$

$$k^n = \left(\frac{1}{k_B T}\right)^{3/2}$$

$$\text{Therefore } \frac{\Gamma n}{k^n} = \frac{\frac{1}{2} \sqrt{\Pi}}{\left(\frac{1}{k_B T}\right)^{3/2}} = \frac{1}{2} \sqrt{\Pi} (k_B T)^{3/2}$$

$$P = \frac{\Pi}{2} \left[\frac{8mh^*}{h^2} \right]^{3/2} \exp \frac{-(E_F - E_V)}{k_B T} \frac{1}{2} \sqrt{\Pi} (k_B T)^{3/2}$$

$$P = \frac{(8)^{3/2}}{4} \left[\frac{mh^* \Pi k_B T}{h^2} \right]^{3/2} \exp \frac{-(E_F - E_V)}{k_B T}$$

$$P = 2 \left[\frac{2mh^* \Pi k_B T}{h^2} \right]^{3/2} \exp \frac{-(E_F - E_V)}{k_B T}$$

$$P = N_V \exp \frac{-(E_F - E_V)}{k_B T} \quad \text{since } N_V = 2 \left[\frac{2mh^* \Pi k_B T}{h^2} \right]^{3/2} \text{pseudo constant}$$

Position of fermi energy level

We know that in intrinsic semiconductor the number of electrons always equal to number of holes

It can be representing $n=p$

we know that

$$N_c \exp \frac{-(E_C - E_F)}{k_B T} = N_V \exp \frac{-(E_F - E_V)}{k_B T}$$

$$\frac{N_V}{N_c} = \frac{\exp \frac{-(E_C - E_F)}{k_B T}}{\exp \frac{-(E_F - E_V)}{k_B T}}$$

$$\frac{N_V}{N_c} = \exp \left(\frac{-E_C - E_V + 2E_F}{k_B T} \right)$$

log on both side

$$\log \frac{N_V}{N_c} = \left(\frac{-E_C - E_V + 2E_F}{k_B T} \right)$$

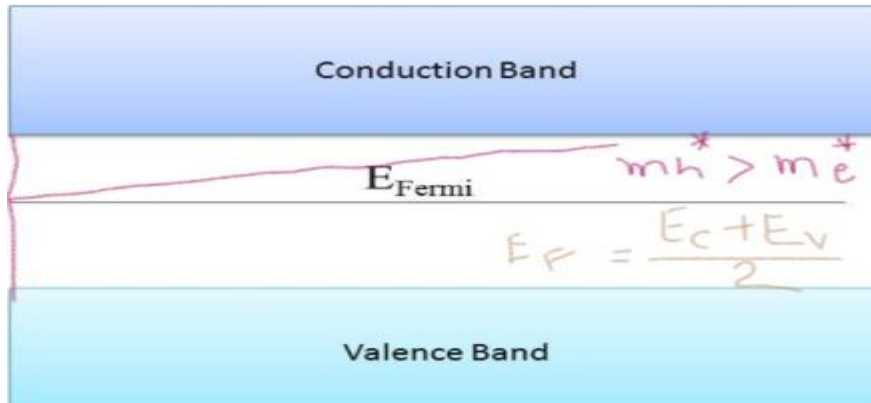
$$2E_F = k_B T \log \left(\frac{N_V}{N_c} \right) + E_C + E_V$$

$$E_F = \frac{k_B T}{2} \log \left(\frac{N_V}{N_C} \right) + \frac{E_C + E_V}{2}$$

$$E_F = \frac{3k_B T}{4} \log \left(\frac{m_h^*}{m_e^*} \right) + \frac{E_C + E_V}{2}$$

If $m_h^* = m_e^*$

$$E_F = \frac{E_C + E_V}{2}$$



the position of the Fermi energy level is exactly the midpoint of the two bands it can be represented

$$E_F = \frac{E_C + E_V}{2}$$

As temperature is rises it is shifted towards conduction band since $m_h^* > m_e^*$

Carrier concentration of intrinsic semiconductor

We know that in intrinsic semiconductor the number of electrons always equal to number of holes

It can be representing $n=p=n_i$ here n_i Carrier concentration

$$n \times p = n_i^2 \text{ ----(1)}$$

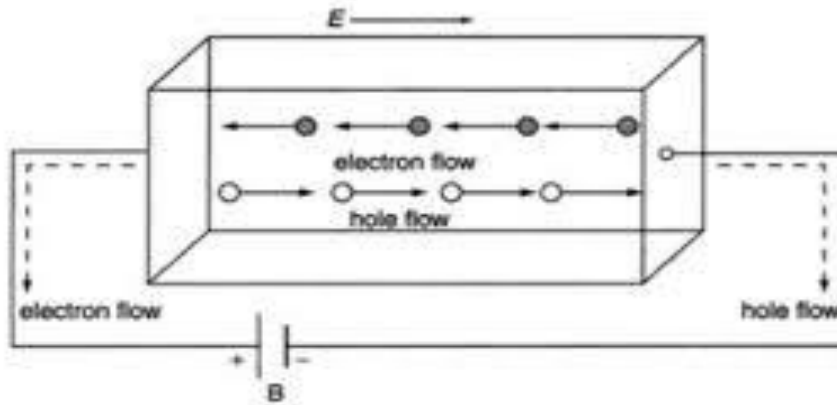
$$2 \left[\frac{2m_e^* \pi k_B T}{h^2} \right]^{3/2} \exp \frac{-(E_C - E_F)}{k_B T} \times 2 \left[\frac{2m_h^* \pi k_B T}{h^2} \right]^{3/2} \exp \frac{-(E_F - E_V)}{k_B T} = n_i^2$$

$$4 \left[\frac{2\pi k_B T}{h^2} \right]^3 (m_e^* m_h^*)^{3/2} \exp \frac{-(E_C - E_V)}{k_B T} = n_i^2$$

$$2^2 \left(\left[\frac{2\pi k_B T}{h^2} \right]^{3/2} \right)^2 \cdot ((m_e^* m_h^*)^{3/4})^2 \cdot \left(\exp\left(\frac{-E_g}{k_B T}\right) \right)^{1/2} = n_i^2$$

$$2 \left[\frac{2\pi k_B T}{h^2} \right]^{3/2} (m_e^* m_h^*)^{3/4} e^{\left(\frac{-E_g}{2k_B T}\right)} = n_i$$

Electrical conductivity of intrinsic semiconductor



Let us consider rectangular bar of semiconductor connecting a battery as shown in above fig

The electrical conductivity due to electrons

$$\sigma_e = n_e u_e$$

The electrical conductivity due to holes

$$\sigma_h = p_e u_h$$

So total The electrical conductivity of intrinsic semiconductor is

$$\sigma_i = \sigma_e + \sigma_h$$

$$\sigma_i = n_e u_e + p_e u_h$$

We know that $n = p = n_i$

$$\sigma_i = n_i u_e + n_i u_h$$

$$\sigma_i = n_i e(u_e + u_h) \text{-----(1)}$$

Mobility of electrons and holes are varying with temperature so it can be written as

$$\text{For electrons } u_e = \frac{\alpha}{T^{3/2}}$$

$$\text{For holes } u_h = \frac{\beta}{T^{3/2}}$$

$$\begin{aligned} (u_e + u_h) &= (\alpha + \beta) T^{-3/2} \\ &= (\gamma) T^{-3/2} \text{-----(2)} \end{aligned}$$

From 1 and 2

$$\sigma_i = n_i e(\gamma) T^{-3/2} \text{-----(3)}$$

$$\text{We know that } n_i = 2 \left[\frac{2\pi k_B T}{h^2} \right]^{3/2} (m_e^* m_h^*)^{3/4} \exp\left(\frac{-E_g}{2k_B T}\right)$$

$$\sigma_i = 2 \left[\frac{2\pi k_B T}{h^2} \right]^{3/2} (m_e^* m_h^*)^{3/4} \exp\left(\frac{-E_g}{2k_B T}\right) e(\gamma) T^{-3/2}$$

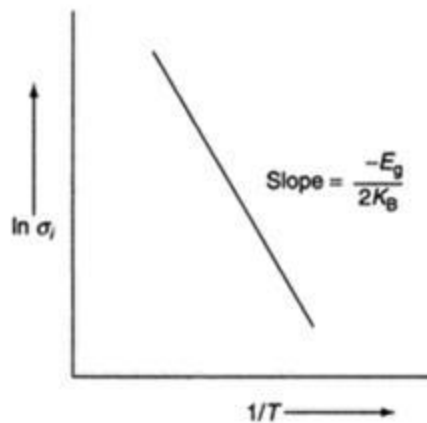
$$\sigma_i = 2 \left[\frac{2\pi k_B}{h^2} \right]^{3/2} (m_e^* m_h^*)^{3/4} T^{3/2} \exp\left(\frac{-E_g}{2k_B T}\right) e(\gamma) T^{-3/2}$$

$$\sigma_i = A T^{3/2} \exp\left(\frac{-E_g}{2k_B T}\right) e(\gamma) T^{-3/2} \text{ since } A = 2 \left[\frac{2\pi k_B}{h^2} \right]^{3/2} (m_e^* m_h^*)^{3/4}$$

$$\sigma_i = A \exp\left(\frac{-E_g}{2k_B T}\right) e(\gamma) \text{ since } A e \gamma = c$$

$$\sigma_i = c \exp\left(\frac{-E_g}{2k_B T}\right) \text{ taking log on both side}$$

$$\ln \sigma_i = \left(\frac{-E_g}{2k_B}\right) \frac{1}{T} + \ln c$$



A graph can be drawn between $\ln \sigma$ versus $1/T$, the resulting graph is a straight line. from above graph it is observed that conductivity increases with increase temperature.

Position of fermi energy level -n-type semiconductor

We know that $n + N_a^- = p + N_d^+$

Formation of n-type material $n = N_d^+$

Formation of p-type material $p = N_a^-$

Here n is $n = 2 \left[\frac{2me^* \pi k_B T}{h^2} \right]^{3/2} \exp \frac{-(E_C - E_F)}{k_B T}$

$$n = 2 \left[\frac{2me^* \pi k_B T}{h^2} \right]^{3/2} \exp \frac{(E_F - E_C)}{k_B T}$$

$$N_d^+ = N_d f(E) = N_d \frac{1}{1 + \exp \left(\frac{E_F - E_d}{k_B T} \right)} = N_d \exp \left(\frac{E_d - E_F}{k_B T} \right)$$

Formation of n-type material $n = N_d^+$

$$2\left[\frac{2me^*\pi k_B T}{h^2}\right]^{3/2} \exp\left(\frac{E_F - E_c}{k_B T}\right) = N_d \exp\left(\frac{E_d - E_F}{k_B T}\right)$$

Taking log on both side of above eqe and we get

$$\frac{(E_F - E_c)}{k_B T} - \left(\frac{E_d - E_F}{k_B T}\right) = \log N_d - \log 2\left[\frac{2me^*\pi k_B T}{h^2}\right]^{3/2}$$

$$\frac{(E_F - E_c)}{k_B T} - \left(\frac{E_d - E_F}{k_B T}\right) = \log \frac{N_d}{2\left[\frac{2me^*\pi k_B T}{h^2}\right]^{3/2}}$$

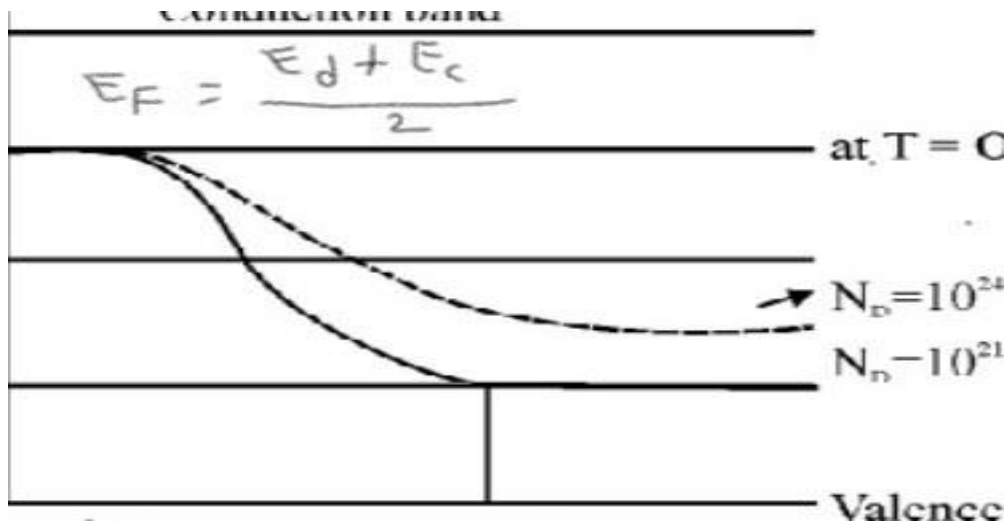
$$E_F = \frac{E_d + E_c}{2} + \frac{k_B T}{2} \log \frac{N_d}{2\left[\frac{2me^*\pi k_B T}{h^2}\right]^{3/2}}$$

If $T=0$

$$E_F = \frac{E_d + E_c}{2}$$

Variation of fermi level with temperature(n-type)

As the Temperature is increased more and more Donor atoms are ionized. for a particular temperature, all the donor atoms are ionized. further increase in temperature results in the generation of electron-hole pairs due to breaking of covalent bond and metals tends to behave like intrinsic manner. Fermi level is gradually decreasing towards intrinsic Fermi level



Position of fermi energy level -p-type semiconductor

We know that $n + N_a^- = p + N_d^+$

Formation of n-type material $n = N_d^+$

Formation of p-type material $p = N_a^-$

$$P = 2 \left[\frac{2m h^* \pi k_B T}{h^2} \right]^{3/2} \exp \frac{-(E_F - E_V)}{k_B T}$$

$$P = 2 \left[\frac{2m h^* \pi k_B T}{h^2} \right]^{3/2} \exp \frac{(E_V - E_F)}{k_B T}$$

$$N_a^- = N_a f(E) = N_a \frac{1}{1 + \exp \left(\frac{E - E_F}{k_B T} \right)} = N_a \exp \left(\frac{E_F - E_a}{k_B T} \right)$$

Formation of P-type material $p = N_a^-$

$$2 \left[\frac{2m h^* \pi k_B T}{h^2} \right]^{3/2} \exp \frac{(E_V - E_F)}{k_B T} = N_a \exp \left(\frac{E_F - E_a}{k_B T} \right)$$

Taking log on both side of above eqe and we get

$$\frac{(E_V - E_F)}{k_B T} - \left(\frac{E_F - E_a}{k_B T} \right) = \log N_d - \log 2 \left[\frac{2m h^* \pi k_B T}{h^2} \right]^{3/2}$$

$$\frac{(E_V - E_F)}{k_B T} - \left(\frac{E_F - E_a}{k_B T} \right) = \log \frac{N_a}{2 \left[\frac{2m h^* \pi k_B T}{h^2} \right]^{3/2}}$$

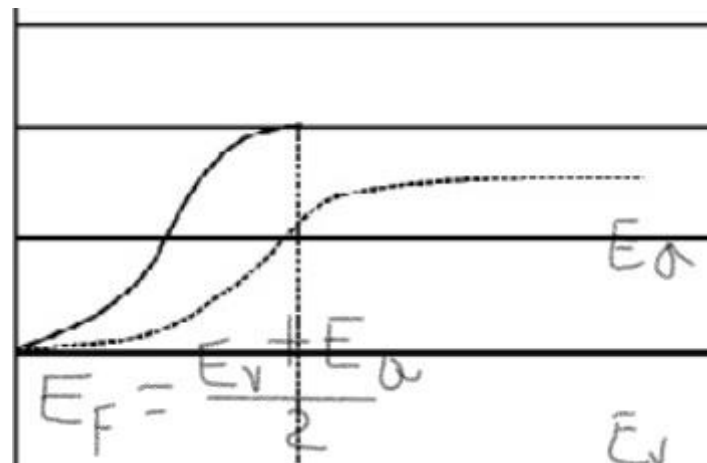
$$E_F = \frac{E_V + E_a}{2} - \frac{k_B T}{2} \log \frac{N_a}{2 \left[\frac{2m h^* \pi k_B T}{h^2} \right]^{3/2}}$$

If $T=0$

$$E_F = \frac{E_V + E_a}{2}$$

Variation of fermi level with temperature(p-type)

As the Temperature is increased more and more acceptor atoms are ionized. for a particular temperature, all the acceptor atoms are ionized. further increase in temperature results in the generation of electron-hole pairs due to breaking of covalent bond and metals tends to behave like intrinsic manner. Fermi level is gradually decreasing towards intrinsic Fermi level



Density of electrons in the conduction band (n-type)

We know that conduction of electrons

$$n = N_c \exp \frac{-(E_C - E_F)}{k_B T}$$

$$n = 2 \left[\frac{2me^* \pi k_B T}{h^2} \right]^{3/2} \exp \frac{-(E_C - E_F)}{k_B T}$$

$$n = 2 \left[\frac{2me^* \pi k_B T}{h^2} \right]^{3/2} \exp \frac{(E_F - E_C)}{k_B T} \text{ ---- (1)}$$

$$E_F = \frac{E_d + E_C}{2} + \frac{k_B T}{2} \log \frac{N_d}{2 \left[\frac{2me^* \pi k_B T}{h^2} \right]^{3/2}} \text{ --- (2)}$$

Now eqe 2 in 1

$$n = 2 \left[\frac{2me^* \pi k_B T}{h^2} \right]^{3/2} \exp \left(\frac{\frac{E_d + E_C}{2} + \frac{k_B T}{2} \log \frac{N_d}{2 \left[\frac{2me^* \pi k_B T}{h^2} \right]^{3/2}} - E_C}{k_B T} \right)$$

$$n = 2 \left[\frac{2me^* \pi k_B T}{h^2} \right]^{3/2} \exp \left(\frac{E_d - E_C}{2k_B T} + \frac{1}{2} \cdot \log \frac{N_d}{2 \left[\frac{2me^* \pi k_B T}{h^2} \right]^{3/2}} \right)$$

$$\text{since } \frac{1}{2} \log x = \log x^{\frac{1}{2}}$$

$$n = 2 \left[\frac{2me^* \pi k_B T}{h^2} \right]^{3/2} \exp \left(\frac{E_d - E_C}{2k_B T} + \log \frac{N_d^{\frac{1}{2}}}{2^{\frac{1}{2}} \left[\frac{2me^* \pi k_B T}{h^2} \right]^{3/4}} \right)$$

since $\exp(a+b) = \exp(a) \cdot \exp(b)$

$$n = 2 \left[\frac{2me^* \pi k_B T}{h^2} \right]^{3/2} \exp \left(\left(\frac{E_d - E_C}{2k_B T} \right) \left(\frac{N_d^{\frac{1}{2}}}{2^{\frac{1}{2}} \left[\frac{2me^* \pi k_B T}{h^2} \right]^{3/4}} \right) \right)$$

$$n = 2 \left[\frac{2me^* \pi k_B T}{h^2} \right]^{3/2} \exp \left(\left(\frac{E_d - E_C}{2k_B T} \right) \left(\frac{N_d}{2} \right)^{\frac{1}{2}} \frac{1}{\left[\frac{2me^* \pi k_B T}{h^2} \right]^{3/4}} \right)$$

$$n = (2N_d) \cdot \frac{1}{2} \left[\frac{2m_e^* \pi k_B T}{h^2} \right]^{3/4} \exp \left(\frac{E_d - E_c}{2k_B T} \right)$$

Density of holes in the valance band (p-type)

We know that conduction of holes

$$p = N_V \exp \frac{-(E_F - E_V)}{k_B T}$$

$$p = 2 \left[\frac{2m_h^* \pi k_B T}{h^2} \right]^{3/2} \exp \frac{-(E_F - E_V)}{k_B T}$$

$$p = 2 \left[\frac{2m_h^* \pi k_B T}{h^2} \right]^{3/2} \exp \frac{(E_V - E_F)}{k_B T} \text{-----(1)}$$

$$E_F = \frac{E_V + E_a}{2} - \frac{k_B T}{2} \log \frac{N_a}{2 \left[\frac{2m_h^* \pi k_B T}{h^2} \right]^{3/2}} \text{-----(2)}$$

Now eqe 2 in 1

$$p = 2 \left[\frac{2m_h^* \pi k_B T}{h^2} \right]^{3/2} \exp \left(\frac{E_V - \frac{E_V + E_a}{2} - \frac{k_B T}{2} \log \frac{N_a}{2 \left[\frac{2m_h^* \pi k_B T}{h^2} \right]^{3/2}}}{k_B T} \right)$$

$$p = 2 \left[\frac{2m_h^* \pi k_B T}{h^2} \right]^{3/2} \exp \left(\frac{E_V - E_a}{2k_B T} + \log \frac{N_a^{1/2}}{2^{1/2} \left[\frac{2m_h^* \pi k_B T}{h^2} \right]^{3/4}} \right)$$

since $\exp(a+b) = \exp(a) \cdot \exp(b)$

$$p = 2 \left[\frac{2m_h^* \pi k_B T}{h^2} \right]^{3/2} \exp \left(\left(\frac{E_V - E_a}{2k_B T} \right) \left(\frac{N_a^{1/2}}{2^{1/2} \left[\frac{2m_h^* \pi k_B T}{h^2} \right]^{3/4}} \right) \right)$$

$$p = 2 \left[\frac{2m h^* \pi k_B T}{h^2} \right]^{3/2} \exp \left(\left(\frac{E_V - E_a}{2k_B T} \right) \left(\frac{N_a}{2} \right)^{\frac{1}{2}} \frac{1}{\left[\frac{2m h^* \pi k_B T}{h^2} \right]^{3/4}} \right)$$

$$p = (2N_a)^{\frac{1}{2}} \left[\frac{2m h^* \pi k_B T}{h^2} \right]^{3/4} \exp \left(\frac{E_V - E_a}{2k_B T} \right)$$

Drift and Diffusion currents

Drift currents We know that the electric current due to moments of charge carriers in the metal, in the semiconductor the charge Carriers are electrons in conduction band the holes in the valance band when the absence of external field they randomly in all directions and hence the net current in any direction is zero when an electric field E is applied across the semiconductor every charge carries experiences a force due to electric field and drifts in the direction of the force. thus, the charge Carriers acquire an average velocity which is called the Drift velocity and it gives rise to the drift current. the total current due to hole and electrons in the presence of the applied electric field is called the drift current

let **n, p** the concentration of electrons and holes and V_{dn} and V_{dp} are Drift velocity of electrons and holes and μ_e and μ_h are mobility of electrons and holes

then it can be written as

$$V_{dn} = n_e u_e E$$

$$V_{dp} = p_e u_h E$$

We know the current density $J_{drift} = n_e u_e E + p_e u_h E$
 $= e E (n u_e + p u_h)$

We know that $n = p = n_i$

$$J_{drift} = n_i e E (u_e + u_h)$$

We know that $j = \sigma E$

$$\sigma E = n_i e E (u_e + u_h)$$

$$\sigma = n_i e (u_e + u_h)$$

Diffusion currents

In addition to the drift current, there is also another current called diffusion current. The diffusion current is due to the flow of charge carriers when there is a non-uniform concentration of charge carriers in semiconductors. The excess carriers move from the region of higher density to the region of lower density, tending to produce a uniform distribution. The motion of charge carriers from the high concentration area to the region of lower concentration area leads to a current called diffusion current. In general, the concentration of carriers varies with distance in a semiconductor, this constitutes a concentration gradient. The diffusion current is proportional to the concentration gradient.

According to Fick's law, the flux of the particles moving from high concentration area to low concentration area is proportional to the concentration gradient, but opposite to the field direction.

It is mathematically written as

$$\begin{aligned} & -Dn \frac{\partial n}{\partial x} \quad \text{for electron} \quad -Dn \frac{\partial n}{\partial x} \cdot (-e) \\ & -Dp \frac{\partial p}{\partial x} \quad \text{for hole} \quad -Dp \frac{\partial p}{\partial x} \cdot e \\ J_{\text{diffusion}} &= -eDp \frac{\partial p}{\partial x} + eDn \frac{\partial n}{\partial x} \end{aligned}$$

Thus, in semiconductor the total current density is due to drift and diffusion of carriers and hence the total current flow in a semiconductor in one dimension is expressed as

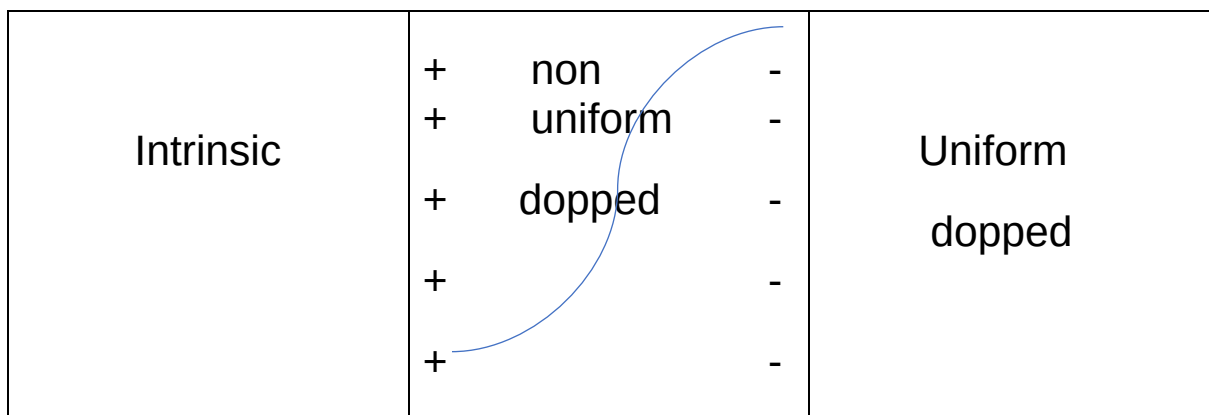
$$\begin{aligned} J &= J_{\text{drift}} + J_{\text{diffusion}} \\ J &= eE(nu_e + pu_h) - eDp \frac{\partial p}{\partial x} + eDn \frac{\partial n}{\partial x} \end{aligned}$$

$$\left(n u_e E + D n \frac{\partial n}{\partial x} \right) + \left(p u_h E - D p \frac{\partial p}{\partial x} \right)$$

Einstein's equation

Einstein equations give the relation between drift, diffusion, and mobility of electrons.

let us consider a semiconductor rod that is intrinsic up to $x=0$ and non-uniform doped in between $x = 0$ to $x = l$ and uniform doped after $x=l$ as shown in the below figure.



$X=0$

$x=l$

the electrons are drifting from $x = l$ to $x = 0$ and accommodated at $x = 0$ layer. here $x = l$ acts as electrically positive due to ionized donor atoms in between $x = 0$ to $x = l$, and they get drift and diffusion current

we know that diffusion current due to electrons

$$e D n \frac{\partial \Delta n}{\partial x} \text{-----(1)}$$

Similarly drift current due to electrons

$$\Delta n_e u_n E \text{-----(2)}$$

From eqe 1 and 2 we get

$$e Dn \frac{\partial \Delta n}{\partial x} = \Delta n_e u_n E$$

$$\Delta n_e E = \frac{Dn}{u_n} e \frac{\partial \Delta n}{\partial x} \text{-----(3)}$$

The force F on excess carriers restoring is given by the product of excess charge and electric field

$$\text{Therefore } \Delta n_e E = F$$

$$F = \frac{Dn}{u_n} e \frac{\partial \Delta n}{\partial x} \text{-----(4)}$$

According to kinetic theory of gas this force also depends on the thermal energy of excess carriers.

If the molecular concentration of a gas increase by (dn), the gas pressure increases by (dn) $k_B T$. the force corresponding to the pressure gradient is given by

$$F = K_B T \cdot \frac{\partial \Delta n}{\partial x} \text{-----(5)}$$

From eqe 4 and 5

$$K_B T \cdot \frac{\partial \Delta n}{\partial x} = \frac{Dn}{u_n} e \frac{\partial \Delta n}{\partial x}$$

$$\frac{Dn}{u_n} = \frac{K_B T}{e} \text{-----(6)}$$

Similarly for holes

$$\frac{Dp}{u_p} = \frac{K_B T}{e} \text{-----(7)}$$

From eqe 6 and 7

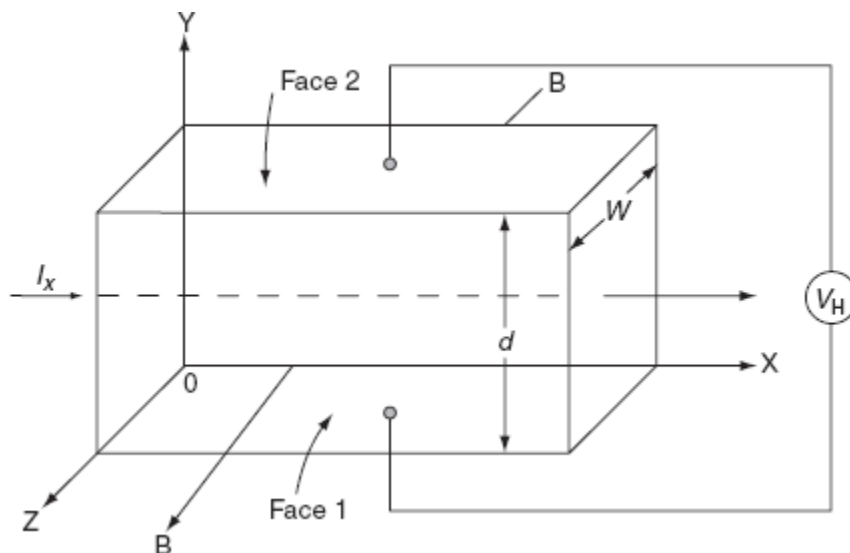
$$\frac{Dn}{u_n} = \frac{Dp}{u_p} \text{-----(8)}$$

$$\frac{Dn}{Dp} = \frac{u_n}{u_p} \text{-----(9)}$$

The above eqs 6,7,9 are Einstein equations

State and explain hall effect, hall coefficients and applications

When a slab of semiconductor carrying current is placed in a transverse magnetic field. a potential difference is produced in the direction to both current magnetic fields. this phenomenon is called hall effect and the generated voltage is called hall voltage. this effect was first explained by E.H. Hall in 1879.



Consider a slab of semiconductor in which a current I is flowing in the positive X- direction as shown in below fig. let a magnetic field B applied along Z- directions then the electrons experience a Lorentz force is given by

$$F_L = -B_e v_d \text{-----}(1)$$

Here B is magnetic field

e is charge of electron

v_d is drift velocity of electrons

Here the bottom surface of the semiconductor is negatively charged and top surface of the semiconductor is positively charged and hence potential V_H is called hall voltage appears between the upper and lower surface of the semiconductor which established an electric field E_H is called hall electric field. the hall electric field is exerting an upward force F_H in the electron is given by

$$F_H = -eE_H \text{-----}(2)$$

Here E_H is hall electric field

e is charge of electron

From eqe 1 and 2

$$F_L = F_H$$

$$-B_e v_d = -eE_H$$

$$E_H = B.v_d \text{-----}(3)$$

For n-type material the majority charge carries are electrons and the current density

$$J = -nev_d$$

Here n is concentration of charge carriers

$$v_d = \frac{-J}{ne} \text{-----}(4)$$

From eqe 3 and 4

$$E_H = \frac{-JB}{ne} \text{-----}(5)$$

For a given semiconductor *the* hall electric field E_H is proportional to the magnetic induction and current density it can be written as

$$E_H \propto JB$$

$$E_H = R_H JB \text{ since } R_H \text{ hall coefficient} \quad (6)$$

From eqe 5 and 6

$$\frac{-JB}{ne} = R_H JB$$

$$R_H = \frac{-1}{ne}$$

For p-type material the majority charge carries are holes then the hall coefficient is

$$R_H = \frac{1}{pe}$$

Determination of hall coefficient R_H

The hall electric field per unit current density per unit magnetic induction is called hall coefficient R_H

If W is width of the sample across which hall voltage V_H is measured

$$E_H = \frac{V_H}{W}$$

$$\text{We know that } E_H = R_H JB$$

$$\text{from above two eqes } \frac{V_H}{W} = R_H JB$$

$$R_H = \frac{V_H}{JBW}$$

$$V_H = R_H JBW$$

If t is the thickness of the sample, then its cross section is **Wt**

And current density is

$$j = \frac{I}{Wt}$$

$$V_H = \frac{R_H IB}{t}$$

$$R_H = \frac{V_H t}{IB}$$

Applications of hall effect

- 1.the sine of charge carriers can be determined
- 2.the carrier density can be estimated
- 3.the mobility of charge carrier can be measured directly
- 4.It can be used to determine whether the given material is conductor or insulator or semiconductor
5. the magnetic field can be measured by knowing the values of hall voltage and hall effect.

Problem-1

Hall coefficients experiment of current(I) 0.25 amp is sent through a metal strip having thickness (t)0.2mm and width(w) is 5mm hall voltage (V_H)is 0.15mv and magnetic field(B) is 2000 tesla calculate current concentration (n)

Formula $E_H = R_H JB$

$$E_H = \frac{1}{ne} \frac{I}{A} B \quad A=wt$$

$$\frac{V_H}{W} = \frac{1}{ne} \frac{I}{wt} B$$

$$n = \frac{IB}{ewt} \frac{W}{V_H}$$

PROBLEM-2

Hall coefficient of semiconductor R_H is 3.66×10^{-4} its resistivity is 8.93×10^{-3} find mobility(u) and concentration of electrons (n)

Formula-1

$$R_H = \frac{1}{ne} \quad n = \frac{1}{R_H e}$$

Formula-2

$$\sigma = neu \quad \sigma = \frac{1}{\rho} \quad u = \frac{1}{\rho ne}$$

Problem-3

Calculate the electrical conductivity and resistivity of intrinsic crystal. the mobility of electron and hole of intrinsic crystal at room temp are 0.36 and 0.17 m^2/V then electron and hole density is $2.5 \times 10^{19} \text{ m}^{-3}$

Formula $\sigma = n_i e (u_e + u_h)$

Problem-4

Find the diffusion coefficient m^2/V of electron at 300 K if u_e is 0.19 m^2/V

Formula $D_n = \frac{u_e K_B T}{e} \quad K_B = 1.38 \times 10^{-23} \text{ J/K}$